

Complex Numbers

Question1

If a complex number $z = x + iy$ represents a point P on the argand plane and $\arg\left(\frac{z-3+2i}{z+2-3i}\right) = \frac{\pi}{4}$, then the locus of P is a

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Options:

A.

circle with the line $x + y = 12$ as its diameter

B.

circle with radius $\sqrt{11}$

C.

circle with the line $x - y = 6$ as its diameter

D.

circle with radius 5

Answer: D

Solution:

$$\because \arg\left(\frac{Z-3+2i}{Z+2-3i}\right) = \frac{\pi}{4} \text{ let } A = 3 - 2i, B = -2 + 3i$$

this means, the angle between $Z - A$ and $Z - B$ is $\pi/4$ i.e., the point Z subtends an angle of $\pi/4$ at the fixed point Z lies on circle that subtends angle $\frac{\pi}{4}$ at the chord joining points A and B .

Now, distance between A and $B = |A - B|$

$$\begin{aligned} &= |(3 - 2i) - (-2 + 3i)| = |5 - 5i| \\ &= 5\sqrt{2} \end{aligned}$$

therefore,

$$\begin{aligned} \text{Radius}(R) &= \frac{AB}{2 \sin \theta} = \frac{5\sqrt{2}}{2 \sin \pi/4} \\ &= \frac{5\sqrt{2}}{2 \times \frac{1}{\sqrt{2}}} = 5 \end{aligned}$$

Hence, the locus of P is a circle with radius 5 .

Question2

By taking $\sqrt{a \pm ib} = x \pm iy, x > 0$, if we get $\frac{\sqrt{21+12\sqrt{2}i}}{\sqrt{21-12\sqrt{2}i}} = a + ib$, then $\frac{b}{a} =$

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Options:

A.

$$\frac{4\sqrt{2}}{7}$$

B.

$$\frac{12\sqrt{2}}{17}$$

C.

$$\frac{4\sqrt{3}}{7}$$

D.

$$\frac{12\sqrt{3}}{17}$$

Answer: A

Solution:

$$\text{Let } \sqrt{\frac{A}{B}} = \sqrt{\frac{21+12\sqrt{2}i}{21-12\sqrt{2}i}}$$

Now,

$$\frac{21+12\sqrt{2}i}{21-12\sqrt{2}i} = \frac{21+12\sqrt{2}i}{21-12\sqrt{2}i} \times \frac{21+12\sqrt{2}i}{21+12\sqrt{2}i}$$

$$= \frac{(21 + 12\sqrt{2}i)^2}{(21)^2 - (12\sqrt{2}i)^2}$$

$$= \frac{153 + 504\sqrt{2}i}{729}$$

$$= \frac{153}{729} + \frac{504\sqrt{2}}{729}i$$

$$\text{then, } \sqrt{\frac{153}{729} + \frac{504\sqrt{2}i}{729}} = a + ib$$

Squaring both sides, we get

$$\frac{153}{729} + \frac{504\sqrt{2}}{729}i = a^2 - b^2 + (2ab)i$$

$$\text{So, } a^2 - b^2 = \frac{153}{729} \text{ and } ab = \frac{252\sqrt{2}}{729}$$

$$\Rightarrow a^2 - b^2 = \frac{17}{81} \text{ and } ab = \frac{28\sqrt{2}}{81}$$

let $b = ak$

$$\Rightarrow b/a = k$$

$$\text{So, } a^2(1 - k^2) = \frac{17}{81} \text{ and } a^2k = \frac{28\sqrt{2}}{81}$$

$$\text{So, } \frac{k}{1-k^2} = \frac{28\sqrt{2}}{81} \times \frac{81}{17} = \frac{28\sqrt{2}}{17}$$

From the options, $k = \frac{4\sqrt{2}}{7}$ satisfy the equation.

Question3

Two values of $(-8 - 8\sqrt{3}i)^{1/4}$ are

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Options:

A.

$$\sqrt{3} - i, -1 - \sqrt{3}i$$

B.

$$\sqrt{3} + i, 1 + \sqrt{3}i$$

C.

$$-\sqrt{3} + i, \sqrt{3} + i$$

D.

$$1 - \sqrt{3}i, \sqrt{3} + i$$

Answer: A

Solution:

We have, $(-8 - 8\sqrt{3}i)^{1/4}$

$$\text{let, } z = -8 - 8\sqrt{3}i$$

$$|z| = \sqrt{(-8)^2 + (-8\sqrt{3})^2} = 16$$

$$\text{and } \theta = \tan^{-1} \left(\frac{-8\sqrt{3}}{-8} \right) = \pi/3$$

so, but the complex number lies in 3rd quadrant

$$\text{So, } \theta = \pi + \pi/3 = 4\pi/3$$

$$\text{then, } z = 16e^{i4\pi/3}$$

$$\text{So, } z^{1/4} = (16)^{1/4} \cdot e^{i(\frac{4\pi}{3} + 2\pi k)/4}$$

$$k = 0, 1, 2, 3, \dots$$

for $k = 0$,

$$z^{1/4} = 2e^{i(\frac{4\pi}{3})/4} = 2e^{i\frac{\pi}{3}} = 1 + i\sqrt{3}$$

for $k = 1$,

$$\begin{aligned} z^{1/4} &= 2e^{i(\frac{4\pi}{3}+2\pi)/4} = 2e^{i(\frac{\pi}{3}+\frac{\pi}{2})} \\ &= -\sqrt{3} + i \end{aligned}$$

for $k = 2$,

$$\begin{aligned} z^{1/4} &= 2e^{i(\frac{4\pi}{3}+4\pi)/4} = 2e^{i(\pi+\frac{\pi}{3})} \\ &= -1 - i\sqrt{3} \end{aligned}$$

for $k = 3$,

$$\begin{aligned} z^{1/4} &= 2e^{i(\frac{4\pi}{3}+6\pi)/4} = 2e^{i(\pi/3+3\pi/2)} \\ &= \sqrt{3} - i \end{aligned}$$

Hence, two values are $\sqrt{3} - i, -1 - \sqrt{3}i$

Question4

If $x = 3 - 2\sqrt{3}i$, then $x^4 - 12x^3 + 54x^2 - 108x - 54 =$

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Options:

A.

0

B.

6

C.

-6

D.

9

Answer: D

Solution:

$$\begin{aligned} \text{Given, } x &= 3 - 2\sqrt{3}i \\ \Rightarrow (x - 3) &= -2\sqrt{3}i \end{aligned}$$

Squaring both sides, we get

$$\begin{aligned} (x - 3)^2 &= (-2\sqrt{3}i)^2 \\ \Rightarrow x^2 - 6x + 9 &= -12 \Rightarrow x^2 - 6x = -21 \end{aligned}$$

Again, squaring both sides, we get

$$(x^2 - 6x)^2 = 441$$

$$\Rightarrow x^4 - 12x^3 + 36x^2 = 441$$

$$\text{Now, } x^4 - 12x^3 + 54x^2 - 108x - 54$$

$$= x^4 - 12x^3 + 36x^2 + 18x^2 - 108x - 54$$

$$= 441 + 18(x^2 - 6x - 3)$$

$$= 441 + 18(-21 - 3)$$

$$= 441 - 18(24)$$

$$= 441 - 432 = 9$$

Question5

z_1, z_2, z_3 represent the vertices A, B, C of a $\triangle ABC$ respectively in the argand plane.

If $|z_1 - z_2| = \sqrt{25 - 12\sqrt{3}}$, $\left| \frac{z_1 - z_3}{z_2 - z_3} \right| = \frac{3}{4}$ and $\angle ACB = 30^\circ$, then the area (in sq units) of that triangle is

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Options:

A.

$$\frac{3}{2}$$

B.

$$3$$

C.

$$5$$

D.

$$\frac{5}{2}$$

Answer: B

Solution:

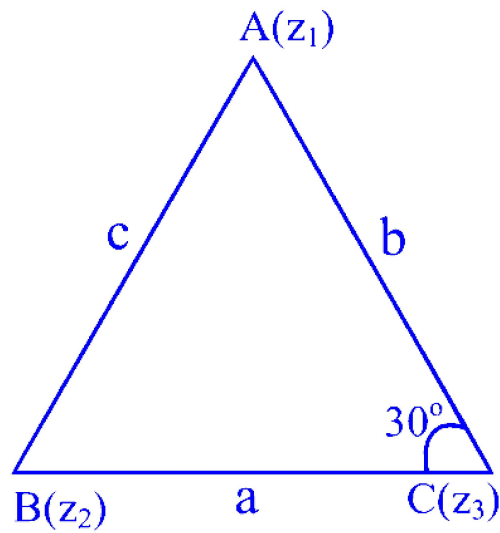
We have,

$$|z_1 - z_2| = \sqrt{25 - 12\sqrt{3}} = c$$

$$\text{and } \frac{|z_1 - z_3|}{|z_2 - z_3|} = \frac{3}{4} = \frac{b}{a} \angle ACB = 30^\circ$$

$$\text{Let } b = 3k, a = 4k$$

$$\therefore \cos C = \frac{b^2 + a^2 - c^2}{2ab}$$



$$\begin{aligned}\frac{\sqrt{3}}{2} &= \frac{9k^2 + 16k^2 - 25 + 12\sqrt{3}}{24k} \\ \Rightarrow 12\sqrt{3}k &= 25k^2 - 25 + 12\sqrt{3} \\ \Rightarrow 25k^2 - 12\sqrt{3}k - 25 + 12\sqrt{3} &= 0 \\ \Rightarrow 25(k^2 - 1) - 12\sqrt{3}(k - 1) &= 0 \\ \Rightarrow (k - 1)(25(k + 1) - 12\sqrt{3}) &= 0 \\ \Rightarrow k = 1 \text{ or } 25(k + 1) = 12\sqrt{3} \\ \Rightarrow k = \frac{12\sqrt{3} - 25}{25} < 0 \\ \therefore k \text{ cannot be negative.} \\ \therefore k = 1 \Rightarrow b = 3, a = 4 \\ \therefore \text{Area of } \triangle ABC &= \frac{1}{2}ab \sin C \\ &= \frac{1}{2} \times 3 \times 4 \times \sin 30^\circ = 3\end{aligned}$$

Question6

The product of the four values of the complex number $(1 + i)^{3/4}$ is

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Options:

A.

$$2(1 + i)$$

B.

$$2(1 - i)$$

C.

$$2^3(1+i)$$

D.

$$2^3(1-i)$$

Answer: B

Solution:

$$\text{Given, } z = (1+i)^{3/4}$$

$$= (\sqrt{2})^{\frac{3}{4}} \left(e^{i\frac{\pi}{4}} \right)^{\frac{3}{4}} = 2^{\frac{3}{8}} \left(e^{i\frac{3\pi}{4}} \right)^{\frac{1}{4}}$$

$$= 2^{3/8} e^{i\frac{3\pi}{4}} \left(2n\pi + \frac{3\pi}{4} \right); n = 0, 1, 2, 3$$

$\therefore$ Product of four values

$$= \left(2^{3/8} \right)^4 e^{i\frac{3\pi}{4}} \left(\frac{3\pi}{4} + 2\pi + \frac{3\pi}{4} + 4\pi + \frac{3\pi}{4} + 6\pi + \frac{3\pi}{4} \right)$$

$$= 2^{3/2} \left(e^{i\frac{15\pi}{4}} \right) = 2^{3/2} \left(e^{i4\pi - \frac{\pi}{4}} \right)$$

$$= 2^{3/2} \left(e^{i4\pi} \right) \cdot e^{i\left(-\frac{\pi}{4}\right)}$$

$$= 2^{3/2} (1) \left(\cos \left(\frac{-\pi}{4} \right) + i \sin \left(\frac{-\pi}{4} \right) \right)$$

$$= 2\sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right) = 2(1-i)$$

Question7

If the point P denotes the complex number $z = x + iy$ in the argand plane and $\frac{z-(2-i)}{z+(1+2i)}$ is purely imaginary number, then the locus of P is

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Options:

A.

a hyperbola not containing the point $(-1, -2)$

B.

an ellipse not containing the point $(-1, -2)$

C.

a parabola not containing the point $(-1, -2)$

D.

a circle not containing the point $(-1, -2)$ and having its centre on the line $x + y + 1 = 0$

Answer: D

Solution:

We note that the quotient $\frac{z-(2-i)}{z+(1+2i)}$ is not defined if $z = -(1+2i) \Leftrightarrow x = -1$ and $y = -2$

Since $x = x + iy$, then

$$\begin{aligned}\frac{z-(2-i)}{z+(1+2i)} &= \frac{x+iy-(2-i)}{x+iy+(1+2i)} \\&= \frac{(x-2)+i(y+1)}{(x+1)+i(y+2)} \\&= \frac{(x-2)+i(y+1)}{(x+1)+i(y+2)} \times \frac{(x+1)-i(y+2)}{(x+1)-i(y+2)} \\&= \frac{(x-2)(x+1)-i(x-2)(y+2)}{+i(y+1)(x+1)+(y+1)(y+2)} \\&= \frac{[(x-2)(x+1)+(y+1)(y+2)]}{(x+1)^2+(y+2)^2} + \frac{i[(x+1)(y+1)-(x-2)(y+2)]}{(x+1)^2+(y+2)^2}\end{aligned}$$

Since, $\frac{z-(2-i)}{z+(1+2i)}$ is purely imaginary,

So real part = 0

$$\begin{aligned}\therefore \frac{(x-2)(x+1)+(y+1)(y+2)}{(x+1)^2+(y+2)} &= 0 \\ \Leftrightarrow x^2+x-2x-2+y^2+2y+y+2 &= 0 \\ x^2+y^2-x+3y &= 0\end{aligned}$$

And $x \neq -1$ and $y \neq -2$

This is the locus of point P . This is a circle.

Question8

If $(\sqrt{3}-i)^n = 2^n, n \in N$, then the least possible value of n is

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Options:

A.

3

B.

4

C.

6

D.

12

Answer: D

Solution:

Given, $(\sqrt{3} - i)^n, n \in N$

$$\begin{aligned}\text{Modulus } |\sqrt{3} - i| &= \sqrt{(\sqrt{3})^2 + (-1)^2} \\ &= \sqrt{3 + 1} = \sqrt{4} = 2\end{aligned}$$

$$\theta = \arg(\sqrt{3} - i) = \tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) = \frac{-\pi}{6}$$

Argument :

$$\theta = \arg(\sqrt{3} - i) = \tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) = \frac{-\pi}{6}$$

So, in polar form,

$$\begin{aligned}\sqrt{3} - i &= 2 \cdot (\cos \theta + i \sin \theta) \\ &= 2 \left[\cos \left(\frac{-\pi}{6} \right) + i \sin \left(\frac{-\pi}{6} \right) \right] \\ &= 2 \left[\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right] \\ \Rightarrow (\sqrt{3} - i)^n &= \left\{ 2 \cdot \left[\cos \left(\frac{\pi}{6} \right) - i \sin \left(\frac{\pi}{6} \right) \right] \right\}^n \\ &= 2^n \cdot \left[\cos \left(\frac{n\pi}{6} \right) - i \sin \left(\frac{n\pi}{6} \right) \right] \quad \dots (i)\end{aligned}$$

$$\text{But given that } (\sqrt{3} - 1)^n = 2^n \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\begin{aligned}2^n \left[\cos \left(\frac{n\pi}{6} \right) - i \sin \left(\frac{n\pi}{6} \right) \right] &= 2^n \\ \Rightarrow \cos \left(\frac{n\pi}{6} \right) - i \sin \left(\frac{n\pi}{6} \right) &= 1 \\ \Rightarrow \cos \left(\frac{n\pi}{6} \right) = 1 \text{ and } \sin \left(-\frac{n\pi}{6} \right) &= 1 \\ \Rightarrow -\frac{n\pi}{6} &= 2k\pi \\ \Rightarrow n &= \frac{-12k\pi}{\pi} = -12k\end{aligned}$$

Since, $n \in N$, for the smallest positive n , let $k = -1$

$$\Rightarrow n = -12(-1) = 12$$

$\therefore$ The least positive value of n is 12 .

Question9

$$(1 + \sqrt{5} + i\sqrt{10 - 2\sqrt{5}})^5 =$$

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Options:

A.

1024

B.

-1024

C.

512

D.

-512

Answer: B

Solution:

Let $z = 1 + \sqrt{5} + i\sqrt{10 - 2\sqrt{5}}$

$$\begin{aligned} \text{So, } |z| &= \sqrt{(1 + \sqrt{5})^2 + (\sqrt{10 - 2\sqrt{5}})^2} \\ &= \sqrt{1 + 5 + 2\sqrt{5} + 10 - 2\sqrt{5}} \\ &= \sqrt{16 + 2\sqrt{5} - 2\sqrt{5}} = \sqrt{16} = 4 \end{aligned}$$

$$\begin{aligned} \text{And } \theta &= \arg(z) = \tan^{-1} \left(\frac{\sqrt{10 - 2\sqrt{5}}}{1 + \sqrt{5}} \right) \\ &= \tan^{-1} \left(\tan \left(\frac{\pi}{5} \right) \right) = \frac{\pi}{5} \end{aligned}$$

So, $z = |z| \cdot (\cos \theta + i \sin \theta)$

$$= 4 \left[\cos \left(\frac{\pi}{5} \right) + i \sin \left(\frac{\pi}{5} \right) \right]$$

$$\begin{aligned} \therefore z^5 &= 4^5 \left[\cos \left(\frac{\pi}{5} \right) + i \sin \left(\frac{\pi}{5} \right) \right]^5 \\ &= 1024 \left[\cos \left(5 \cdot \frac{\pi}{5} \right) + i \sin \left(5 \cdot \frac{\pi}{5} \right) \right] \\ &= 1024 [\cos \pi + i \sin \pi] \\ &= 1024 [-1 + 0] = -1024 \end{aligned}$$

Question10

If z is a complex number such that $\frac{z-1}{z-i}$ is purely imaginary and locus of z represents a circle with centre (α, β) and radius r , then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} =$

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Options:

A.

$4r$

B.

r^2

C.

$2r^2$

D.

$$4r^2$$

Answer: D

Solution:

$$\begin{aligned}\frac{z-1}{z-i} &= \frac{(x+iy)-1}{(x+iy)-i} \\ &= \frac{(x-1)+iy}{x+i(y-1)} \\ &= \frac{(x-1)+iy}{x+i(y-1)} \times \frac{(x-i(y-1))}{(x-i(y-1))} \\ &= \frac{[x(x-1)+y(y-1)]+i[xy-(x-1)(y-1)]}{x^2+(y-1)^2}\end{aligned}$$

$\therefore$ The fraction is purely imaginary.

So, $x^2 - x + y^2 - y = 0$ {real part should be zero}

$$\begin{aligned}\Rightarrow x^2 - x + \frac{1}{4} + y^2 - y + \frac{1}{4} &= \frac{1}{4} + \frac{1}{4} \\ \Rightarrow \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 &= \frac{1}{2} = \left(\frac{1}{\sqrt{2}}\right)^2\end{aligned}$$

Center of circle $(\alpha, \beta) = \left(\frac{1}{2}, \frac{1}{2}\right)$

and Radius $(r) = \frac{1}{\sqrt{2}}$

$$\text{Thus, } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{(1/2)}{(1/2)} + \frac{(1/2)}{(1/2)} = 2$$

and $r = \frac{1}{\sqrt{2}}$

$$\Rightarrow r^2 = \frac{1}{2}$$

$$\text{So, } 4r^2 = 4 \times \frac{1}{2} = 2$$

$$\text{Hence, } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 4r^2$$

Question11

If the least positive integer n satisfying the equation $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^n = -1$ is p and the least positive integer m satisfying the equation $\left(\frac{1-\sqrt{3}i}{1+\sqrt{3}i}\right)^m = \text{cis } \frac{2\pi}{3}$ is q , then $\sqrt{p^2 + q^2} =$

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Options:

A.

5

B.

10

C.

$$\sqrt{13}$$

D.

$$\sqrt{17}$$

Answer: C

Solution:

$$\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^n = -1$$

$$\therefore \text{Modulus of numerator} = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

and modulus of denominator = 2

$$\theta = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

$$\begin{aligned}\therefore \left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^n &= \left(\frac{2 \operatorname{cis}\left(\frac{\pi}{6}\right)}{2 \operatorname{cis}\left(-\frac{\pi}{6}\right)}\right)^n \\ &= \left[\operatorname{cis}\left(\frac{\pi}{6} - \left(-\frac{\pi}{6}\right)\right)\right]^n \\ &= \left[\operatorname{cis}\left(\frac{\pi}{3}\right)\right]^n = \operatorname{cis}\left(\frac{n\pi}{3}\right)\end{aligned}$$

$$\text{and } \operatorname{cis}\left(\frac{n\pi}{3}\right) = -1$$

$$= \operatorname{cis}(\pi)$$

$$\therefore \frac{n\pi}{3} = \pi$$

$$\Rightarrow n = 3$$

thus, $p = 3$

Similarly, second expression

$$\left(\frac{1-\sqrt{3}i}{1+\sqrt{3}i}\right)^m = \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow \left[\frac{2 \operatorname{cis}\left(-\frac{\pi}{3}\right)}{2 \operatorname{cis}\left(\frac{\pi}{3}\right)}\right]^m = \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow \left[\operatorname{cis}\left(-\frac{\pi}{3} - \frac{\pi}{3}\right)\right]^m = \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow \operatorname{cis}\left(-\frac{2\pi}{3} \cdot m\right) = \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$\text{Therefore, } -\frac{2\pi}{3} \cdot m = \frac{2\pi}{3}$$

$$\Rightarrow -2\pi m \equiv 2\pi \pmod{6\pi}$$

$$\Rightarrow -2m \equiv 2 \pmod{6}$$

$$\Rightarrow 2m \equiv -2 \equiv 4 \pmod{6}$$

$$\Rightarrow m \equiv 2 \pmod{3}$$

$\therefore$ Least positive integer = $m = 2$ thus, $q = 2$

$$\text{Hence, } \sqrt{q^2 + p^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

Question12

Sum of the squares of the imaginary roots of the equation $z^8 - 20z^4 + 64 = 0$ is

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Options:

A.

0

B.

-12

C.

-4

D.

-16

Answer: B

Solution:

$$z^8 - 20z^4 + 64 = 0$$

$$\Rightarrow (z^4)^2 - 16z^4 - 4z^4 + 64 = 0$$

$$\Rightarrow (z^4 - 16)(z^4 - 4) = 0$$

$$z^4 = 16 \text{ or } z^4 = 4$$

$$\Rightarrow z = \pm 2 \pm 2i, \pm\sqrt{2}, \pm\sqrt{2}i$$

imaginary roots are $2i, -2i, \sqrt{2}i, -\sqrt{2}i$ sum of square the imaginary roots

$$= (2i)^2 + (-2i)^2 + (\sqrt{2}i)^2 + (-\sqrt{2}i)^2$$

$$= -4 - 4 - 2 - 2 = -12$$

Question13

For any two non-zero complex numbers z_1 and z_2 , if $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, then

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Options:

A.

$$\operatorname{Re}\left(\frac{z_1}{z_2}\right) = 0$$

B.

$$\operatorname{Im}\left(\frac{z_1}{z_2}\right) = 0$$

C.

$$\operatorname{Re}(z_1 z_2) = 0$$

D.

$$\operatorname{Im}(z_1 z_2) = 0$$

Answer: A

Solution:

We are given that:

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$$

For two non-zero complex numbers z_1 and z_2 , we know that:

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2}) = |z_1|^2 + 2\operatorname{Re}(z_1 \overline{z_2}) + |z_2|^2$$

Given that:

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$$

we can equate:

$$|z_1|^2 + 2\operatorname{Re}(z_1 \overline{z_2}) + |z_2|^2 = |z_1|^2 + |z_2|^2$$

Simplifying:

$$2\operatorname{Re}(z_1 \overline{z_2}) = 0$$

Thus, the **real part** of $z_1 \overline{z_2}$ must be zero:

$$\operatorname{Re}(z_1 \overline{z_2}) = 0$$

This means that $z_1 \overline{z_2}$ is purely imaginary, implying that the real part of $\frac{z_1}{z_2}$ is also zero.

Final Answer:

The correct answer is **Option A**, which states:

$$\operatorname{Re}\left(\frac{z_1}{z_2}\right) = 0$$

Question14

If $1, \omega, \omega^2$ are the cube roots of unity, then

$$1\left(2 + \frac{1}{\omega}\right)\left(2 + \frac{1}{\omega^2}\right) + 2\left(3 + \frac{1}{\omega}\right)\left(3 + \frac{1}{\omega^2}\right) + 3\left(4 + \frac{1}{\omega}\right)\left(4 + \frac{1}{\omega^2}\right) + \dots 10 \text{ terms} =$$

Options:

A.

3080

B.

3465

C.

3175

D.

3715

Answer: B**Solution:**

Given, $1, \omega, \omega^2$ are cube roots of unity.

Also, we can write the general term (t_n) of given expression

$$\begin{aligned}
 t_n &= (n-1) \left(n + \frac{1}{\omega} \right) \left(n + \frac{1}{\omega^2} \right) \\
 &= (n-1) (n + \omega^2) (n + \omega) \\
 &= (n-1) (n^2 + n(\omega + \omega^2) + \omega^3) \\
 &= (n-1) (n^2 - n + 1) [\because 1 + \omega + \omega^2 = 0] \\
 &= (n^3 - 2n^2 + 2n - 1) \quad \dots (i)
 \end{aligned}$$

Now, $\sum_{r=1}^n r = \frac{n(n+1)}{2}$,

$$\begin{aligned}
 \sum_{r=1}^n r^2 &= \frac{n(n+1)(2n+1)}{6} \\
 \sum_{r=1}^n r^3 &= \left(\frac{n(n+1)}{2} \right)^2
 \end{aligned}$$

So, we have;

$$\begin{aligned}
 \sum_{n=2}^{11} n &= \left(\sum_{n=2}^{11} n - 1 \right) \\
 &= \frac{11(12)}{2} - 1 = 66 - 1 = 65 \\
 \sum_{n=2}^{11} n^2 &= \left(\sum_{n=1}^{11} n^2 \right) - 1 \\
 &= \frac{11(12)(23)}{6} - 1 = 506 - 1 = 505 \\
 \sum_{n=2}^{11} n^3 &= \left(\sum_{n=1}^{11} n^3 \right) - 1 \\
 &= \left(\frac{11(12)}{2} \right)^2 - 1 = (66)^2 - 1 \\
 &= 4356 - 1 = 4355
 \end{aligned}$$

$$\text{and } \sum_{n=2}^{11} 1 = 10$$

So, final answer

$$\begin{aligned}
 &= 4355 - 2(505) + 2(65) - 10 \\
 &= 4355 - 1010 + 130 - 10 = 3465
 \end{aligned}$$

Question15

$$(1 + \sqrt{3}i)^6 - (\sqrt{3} + i)^6 =$$

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Options:

A.

0

B.

32

C.

64

D.

128

Answer: D

Solution:

$$\begin{aligned}
 &(1 + \sqrt{3}i)^6 - (\sqrt{3} + i)^6 \\
 &= 2^6 \cdot \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^6 - \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^6 2^6 \\
 &= 2^6 \left[\left(\cos \frac{\pi}{3} + \sin \frac{\pi}{3}i\right)^6 - \left(\cos \frac{\pi}{6} + \sin \frac{\pi}{6}i\right)^6 \right]
 \end{aligned}$$

(Using De-Moivre's theorem)

$$\begin{aligned}
 &= 2^6 [(\cos 2\pi + \sin 2\pi i) - (\cos \pi + \sin \pi i)] \\
 &= 64[(1 + 0) - (-1 + 0)] \\
 &= 64(1 + 1) = 128
 \end{aligned}$$

Question16

If $z = x + iy$ and $x^2 + y^2 = 1$, then $\frac{1+x+iy}{1+x-iy} =$

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Options:

A.

$$\bar{z}$$

B.

$$z$$

C.

$$z + 1$$

D.

$$z - 1$$

Answer: B

Solution:

$$\begin{aligned} z &= x + iy \text{ and } x^2 + y^2 = 1 \\ \frac{1+x+iy}{1+x-iy} &= \frac{((1+x)+iy)((1+x)+iy)}{(1+x)^2 - i^2 y^2} \\ &= \frac{(1+x)^2 + i^2 y^2 + 2(1+x)yi}{1+x^2+2x+y^2} \\ &= \frac{1+x^2+2x-y^2+2(1+x)yi}{2+2x} \\ &= \frac{2x^2+2x+2(1+x)yi}{2(1+x)} \\ &= \frac{2x(1+x)+2(1+x)yi}{2(1+x)} \\ &= x+iy = z \end{aligned}$$

Question17

If $x^6 = (\sqrt{3} - i)^5$, then the product of all of its roots is

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Options:

A.

$$2^5(\sqrt{3} + i)$$

B.

$$\frac{2^6}{\sqrt{3}+i}$$

C.

$$2^6(\sqrt{3} - i)$$

D.

$$\frac{2^6}{\sqrt{3}-i}$$

Answer: D

Solution:

$$\begin{aligned}\sqrt{3}-i &= 2\left(\cos\left(-\frac{\pi}{6}\right)+i\sin\left(-\frac{\pi}{6}\right)\right) \\ &= 2e^{-i\pi/6} \\ \Rightarrow (\sqrt{3}-i)^5 &= \left(2e^{-\frac{i\pi}{6}}\right)^5 = 32e^{-\frac{i5\pi}{6}}\end{aligned}$$

The product of the root of $x^6 = c$ is

$$(-1)^{6-1}c = -c$$

The product of the root is $-32e^{-\frac{i5\pi}{6}}$

$$\begin{aligned}&= -32\left(\cos\left(-\frac{5\pi}{6}\right)+i\sin\left(-\frac{5\pi}{6}\right)\right) \\ &= -32\left(-\frac{\sqrt{3}}{2}-\frac{1}{2}i\right) \\ &= 16\sqrt{3}+16i \\ &= \frac{2^6}{\sqrt{3}-i}\end{aligned}$$

Question18

The minimum value of $|z-1|+|z-5|$ is

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Options:

A.

3

B.

5

C.

4

D.

2

Answer: C

Solution:

$$\begin{aligned}
 |z-1| + |z-5| &\geq |z-1-z+5| \\
 &\geq 4 \left[\because |z_1| + |z_2| \geq |z_1 - z_2| \right] \\
 \text{Minimum value} &= 4
 \end{aligned}$$

Question19

If $z = x + iy$ and if the point P in the argand diagram represents z , then the locus of the point P satisfying the equation $2|z - 2 - 3i| = 3|z + i - 2|$ is a circle with centre

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Options:

A.

$$(10, -21)$$

B.

$$\left(2, -\frac{21}{5}\right)$$

C.

$$(-10, 21)$$

D.

$$\left(-2, \frac{21}{5}\right)$$

Answer: B

Solution:

$$z = x + iy$$

$$2|z - 2 - 3i| = 3|z + i - 2|$$

$$\text{Let } z = x + iy$$

$$\begin{aligned}
 2|(x-2) + i(y-3)| \\
 = 3|(x-2) + i(y+1)|
 \end{aligned}$$

Squaring on both sides, we get

$$4x^2 - 16x + 16 + 4y^2 - 24y + 36$$

$$= 9x^2 - 36x + 36 + 9y^2 + 18y + 9$$

$$\Rightarrow 5x^2 + 5y^2 - 20x + 42y - 7 = 0$$

$$\text{Centre} \equiv \left(2, -\frac{21}{5}\right)$$

Question20

If z is a non-real root of $x^7 = 1$, then $1 + 3z + 5z^2 + 7z^3 + 9z^4 + 11z^5 + 13z^6 =$

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Options:

A.

$$\frac{14}{1-z}$$

B.

$$\frac{-14}{1-z}$$

C.

$$\frac{15}{1-z}$$

D.

$$\frac{-15}{1-z}$$

Answer: B

Solution:

$$\begin{aligned}x^7 &= 1 \\ \Rightarrow x^7 - 1 &= 0 \\ \therefore z^7 - 1 &= 0 \\ (z-1)(1+z+z^2+z^3+z^4+z^5+z^6) &= 0 \\ z &\neq 1 \\ 1+z+z^2+z^3+z^4+z^5+z^6 &= 0 \quad \dots (i)\end{aligned}$$

Now,

$$\begin{array}{r}S = 1 + 3z + 5z^2 + 7z^3 + 9z^4 + 11z^5 + 13z^6 \\ zS = z + 3z^2 + 5z^3 + 7z^4 + 9z^5 + 11z^6 + 13z^7 \\ \hline S(1-z) = 1 + 2z + 2z^2 + \dots + 2z^6 - 13z^7 \\ S(1-z) = -14 \Rightarrow S = \frac{-14}{1-z} \quad [z^7 = 1]\end{array}$$

Question 21

If $\cosh 2x = 199$, then $\cot hx =$

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Options:

A.

$$\frac{5}{3\sqrt{11}}$$

B.

$$\frac{5}{6\sqrt{11}}$$

C.

$$\frac{7}{3\sqrt{11}}$$

D.

$$\frac{10}{3\sqrt{11}}$$

Answer: D

Solution:

$$\text{If } \cosh 2x = 199$$

$$\coth x = \sqrt{\frac{1 + \cosh 2x}{\cosh 2x - 1}} = \sqrt{\frac{200}{198}}$$

$$\Rightarrow \coth 2x = \frac{10}{3\sqrt{11}}$$

Question22

If $a = \operatorname{Im} \left(\frac{1+z^2}{2iz} \right)$ and z is any non-zero complex number such that $|z| = 1$, then $a =$

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Options:

A.

$$\operatorname{Re}(z)$$

B.

$$\operatorname{Re}(z) \operatorname{Im}(z)$$

C.

$$-\operatorname{Re}(z)$$

D.

$$\operatorname{Re}(z) + \operatorname{Im}(z)$$

Answer: C

Solution:

We have, $|z| = 1$

$\therefore$ Let $z = \cos \theta + i \sin \theta = e^{i\theta}$

$$1 + z^2 = 1 + \cos 2\theta + i \sin 2\theta$$

$$= 2 \cos^2 \theta + 2i \sin \theta \cos \theta$$

$$= 2 \cos \theta (\cos \theta + i \sin \theta) = 2 \cos \theta \cdot e^{i\theta}$$

$$\text{Now, } \frac{1 + z^2}{2iz} = \frac{2 \cos \theta \cdot e^{i\theta}}{2ie^{i\theta}} = \frac{\cos \theta}{i} = -i \cos \theta$$

$$\text{Now, } \operatorname{Im} \left(\frac{1 + z^2}{2iz} \right) = -\cos \theta = -\operatorname{Re}(z)$$

Question23

If $(3 + 4i)^{2025} = 5^{2023}(x + iy)$, then $\sqrt{x^2 + y^2} =$

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Options:

A.

5

B.

25

C.

125

D.

625

Answer: B

Solution:

$$\text{Given, } (3 + 4i)^{2025} = 5^{2023}(x + iy)$$

Taking modulus on both sides, we get

$$5^{2025} = 5^{2023} \sqrt{x^2 + y^2}$$

$$\therefore \sqrt{x^2 + y^2} = 5^2 = 25$$

Question24

If $\left(\frac{\cos \theta + i \sin \theta}{\sin \theta + i \cos \theta} \right)^{2024} + \left(\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta + i \sin \theta} \right)^{2025} = x + iy$ then the value of $x + y$ at $\theta = \frac{\pi}{2}$ is

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Options:

A.

1

B.

-1

C.

2

D.

2024

Answer: C

Solution:

$$\begin{aligned} \text{We have, } & \left(\frac{\cos \theta + i \sin \theta}{\sin \theta + i \cos \theta} \right)^{2024} + \left(\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta + i \sin \theta} \right)^{2025} \\ &= \left\{ \frac{(\cos \theta + i \sin \theta)}{i(\cos \theta - i \sin \theta)} \right\}^{2024} + \left(\frac{2 \cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right)^{2025} \\ &= \left(\frac{e^{i\theta}}{e^{-i\theta}} \right)^{2024} + \frac{(2 \cos \frac{\theta}{2})^{2025} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})^{2025}}{(2 \sin \frac{\theta}{2})^{2025} (\sin \frac{\theta}{2} + i \cos \frac{\theta}{2})^{2025}} \\ &= (e^{2i\theta})^{2024} + \left(\cot \frac{\theta}{2} \right)^{2025} \frac{(e^{i\frac{\theta}{2}})^{2025}}{i^{2025} (e^{-i\frac{\theta}{2}})^{2025}} \\ &= (e^{2i\theta})^{2024} + \left(\cot \frac{\theta}{2} \right)^{2025} \frac{1}{i} (e^{i\theta})^{2025} \\ &= (\cos 2\theta + i \sin 2\theta)^{2024} - \left(\cot \frac{\theta}{2} \right)^{2025} i (\cos \theta + i \sin \theta)^{2025} \\ \text{Now, put } \theta &= \frac{\pi}{2} \\ &= (-1)^{2024} - i(i)^{2025} = 1 - i^2 = 1 + 1 = 2 \end{aligned}$$

Question 25

If $a \pm ib$ and $b \pm ai$ are the roots of $x^4 - 10x^3 + 50x^2 - 130x + 169 = 0$, then $\frac{a}{b} + \frac{b}{a} =$

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Options:

A.

$$\frac{25}{12}$$

B.

$$\frac{5}{2}$$

C.

$$\frac{13}{6}$$

D.

$$\frac{34}{15}$$

Answer: C

Solution:

Given,

$$x^4 - 10x^3 + 50x^2 - 130x + 169 = 0$$

$$\text{Sum of roots } 2a + 2b = 10 \Rightarrow a + b = 5 \quad \dots (i)$$

$$\text{Product of roots } (a^2 + b^2)(b^2 + a^2) = 169$$

$$a^2 + b^2 = 13 \quad \dots (ii)$$

From, Eq.s (i) and (ii) $ab = 6$

$$\frac{a^2+b^2}{ab} = \frac{13}{6} \Rightarrow \frac{a}{b} + \frac{b}{a} = \frac{13}{6}$$

Question26

$$\text{If } i = \sqrt{-1}, \text{ then } \sum_{n=2}^{30} i^n + \sum_{n=30}^{65} i^{n+3} =$$

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Options:

A.

$$0$$

B.

$$-1$$

C.

$$i$$

D.

$$-1$$

Answer: B

Solution:

We have, $i = \sqrt{-1}$

$$\sum_{n=2}^{30} i^n = i^2 + i^3 + \dots + i^{30} = 0 + i^{30} = 0 - 1$$

$$\sum_{n=30}^{65} i^{n+3} = i^{33} + i^{34} + \dots + i^{68} = 0$$

$$\therefore \sum_{n=2}^{30} i^n + \sum_{n=30}^{65} i^{n+3} = -1 + 0 = -1$$

Question27

If z_1 and z_2 are two of the n th roots of unity such that the line segment joining them subtends at a right angle at the origin, then for a positive integer k , n takes the form

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Options:

A.

$$4k$$

B.

$$4k + 1$$

C.

$$4k + 2$$

D.

$$4k + 3$$

Answer: A

Solution:

z_1 and z_2 are both n th roots of unity. This means each one can be written in the form $e^{\frac{i2\pi r}{n}}$, where r is an integer.

Let $z_1 = e^{\frac{i2\pi r_1}{n}}$ and $z_2 = e^{\frac{i2\pi r_2}{n}}$ for some integers r_1 and r_2 .

If the line segment joining z_1 and z_2 makes a right angle at the origin, this means the points z_1 , z_2 , and the origin form a right-angled triangle with the right angle at the origin.

This can only happen if $\frac{z_1}{z_2}$ is a purely imaginary number. A complex number is purely imaginary when its argument (angle) is $\frac{\pi}{2}$ or $\frac{3\pi}{2}$, or in general, $\frac{\pi}{2} + m\pi$ where m is an integer.

We find: $\frac{z_1}{z_2} = e^{\frac{i2\pi}{n}(r_1 - r_2)}$

For this to be purely imaginary, we need: $\frac{2\pi}{n}(r_1 - r_2) = \frac{\pi}{2} + m\pi$

Divide both sides by π : $\frac{2}{n}(r_1 - r_2) = \frac{2m+1}{2}$

Now solve for n : $n = \frac{4(r_1 - r_2)}{2m+1}$

Since n must be a positive integer, $2m + 1$ must divide evenly into $4(r_1 - r_2)$. The simplest way for this to always work for integer values is when n is a multiple of 4. Therefore, $n = 4k$ for some positive integer k .

Question 28

$$(\sqrt{\sqrt{2} + 1} + i\sqrt{\sqrt{2} - 1})^8 =$$

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Options:

A.

64

B.

$64i$

C.

-64

D.

$-64i$

Answer: C

Solution:

$$\text{Let, } Z = \sqrt{\sqrt{2} + 1} + i\sqrt{\sqrt{2} - 1}$$

$$\Rightarrow |Z| = \sqrt{\sqrt{2} + 1 + \sqrt{2} - 1} = \sqrt{2\sqrt{2}}$$

$$\cos \theta = \frac{\sqrt{\sqrt{2} + 1}}{\sqrt{2\sqrt{2}}}, \sin \theta = \frac{\sqrt{\sqrt{2} - 1}}{\sqrt{2\sqrt{2}}}$$

$$\therefore \cos^2 \theta = \frac{\sqrt{2} + 1}{2\sqrt{2}}, \sin^2 \theta = \frac{\sqrt{2} - 1}{2\sqrt{2}}$$

$$\Rightarrow \frac{1 + \cos 2\theta}{2} = \frac{\sqrt{2} + 1}{2\sqrt{2}}$$

$$\Rightarrow \cos 2\theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 2\theta = \frac{\pi}{4} \text{ (since, real and imaginary part are positive.)}$$

Now,

$$Z^8 = (\sqrt{2\sqrt{2}})^8 \left(\cos \left(8 \times \frac{\pi}{8} \right) + i \sin \left(8 \times \frac{\pi}{8} \right) \right)$$

$$\begin{aligned}
&= (2\sqrt{2})^4 (\cos \pi + i \sin \pi) \\
&= \left(2^{\frac{3}{2}} \times 4\right) (-1 + 0) \\
&= 2^6 (-1) = -64
\end{aligned}$$

Question29

ω is a complex cube root of unity and if z is a complex number satisfying $|z - 1| \leq 2$ and $|\omega^2 z - 1 - \omega| = a$, then the set of possible values of a is

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Options:

- A. $0 \leq a \leq 2$
- B. $\frac{1}{2} \leq a \leq \frac{\sqrt{3}}{2}$
- C. $|\omega| \leq a \leq \frac{\sqrt{3}}{2} + 2$
- D. $0 \leq a \leq 4$

Answer: D

Solution:

We have, $|z - 1| \leq 2$... (i)

and $|\omega^2 z - 1 - \omega| = a$

$$\omega^2 z + \omega^2 = a \quad [\because \omega^2 = -1 - \omega]$$

$$|\omega^2||z + 1| = a$$

$$|z - 1 + 2| = a$$

$$|z - 1| + 2 \geq a$$

From Eq. (i), we get

$$|z - 1| + 2 = 4$$

$$\Rightarrow 0 \leq a \leq 4$$

Question30

If the roots of the equation $z^3 + iz^2 + 2i = 0$ are the vertices of a $\triangle ABC$, then that $\triangle ABC$ is

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Options:

- A. a right angled triangle
- B. an equilateral triangle
- C. an isosceles triangle

D. a right angled isosceles triangle

Answer: C

Solution:

We have, $z^3 + iz^2 + 2i = 0$

One root of $z^3 + iz^2 + 2i = 0$ is i .

$$\begin{aligned} & [\because i^3 + i(i^2) + 2i = -i - i + 2i = 0] \\ & z^3 + iz^2 + 2i = 0 \\ \Rightarrow & (z - i)(z^2 + 2iz - 2) = 0 \\ \Rightarrow & (z - i)(z^2 + 2iz + i^2 + 1 - 2) = 0 \\ \Rightarrow & (z - i)((z + i)^2 - 1) = 0 \\ \Rightarrow & (z - i)(z + (i + 1))(z + (i - 1)) = 0 \\ \Rightarrow & z = i, -i - 1, 1 - i \end{aligned}$$

$\therefore$ The triangle formed by the vertices $A(0, 1)$, $B(-1, -1)$ and $C(1, -1)$.

$$AB = \sqrt{1 + 4} = \sqrt{5} \text{ units}$$

$$BC = 2 \text{ units}$$

$$AC = \sqrt{1 + 4} = \sqrt{5} \text{ units}$$

$\Rightarrow \triangle ABC$ is an isosceles triangle.

$$[\because AB = AC]$$

Since, the non-equal side is less than the equal side, so ABC is not a right-angled triangle.

Question31

(r, θ) denotes $r(\cos \theta + i \sin \theta)$. If $x = (1, \alpha)$, $y = (1, \beta)$, $z = (1, \gamma)$ and $x + y + z = 0$, then $\Sigma \cos(2\alpha - \beta - \gamma)$ is equal to

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Options:

A. 3

B. 0

C. 1

D. -1

Answer: A

Solution:

In the complex plane, any point can be represented in polar form as $(r, \theta) = r(\cos \theta + i \sin \theta)$. Here, $x = (1, \alpha)$, $y = (1, \beta)$, and $z = (1, \gamma)$. Given the condition $x + y + z = 0$, we can express

$$x = \cos \alpha + i \sin \alpha$$

Similarly,

$$y = \cos \beta + i \sin \beta$$

$$z = \cos \gamma + i \sin \gamma$$

These imply:

$$x + y + z = 0 \Rightarrow \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\sin \alpha + \sin \beta + \sin \gamma = 0$$

From the above, we isolate:

$$\cos \alpha + \cos \beta = -\cos \gamma \quad \dots(i)$$

$$\sin \alpha + \sin \beta = -\sin \gamma \quad \dots(ii)$$

Squaring and adding equations (i) and (ii), we get:

$$\begin{aligned} (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 &= \cos^2 \gamma + \sin^2 \gamma \\ \cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta + \sin^2 \alpha + \sin^2 \beta + 2 \sin \alpha \sin \beta &= 1 \end{aligned}$$

This simplifies to:

$$2 \cos(\alpha - \beta) = -1 \quad \Rightarrow \quad \cos(\alpha - \beta) = -\frac{1}{2}$$

By symmetry,

$$\cos(\beta - \gamma) = -\frac{1}{2}$$

and

$$\cos(\gamma - \alpha) = -\frac{1}{2}$$

The corresponding sine values are:

$$\sin(\alpha - \beta) = \sin(\beta - \gamma) = \sin(\gamma - \alpha) = \frac{\sqrt{3}}{2}$$

Consider $\cos(2\alpha - \beta - \gamma)$:

$$\begin{aligned} \cos(2\alpha - \beta - \gamma) &= \cos((\alpha - \beta) - (\gamma - \alpha)) \\ &= \cos(\alpha - \beta) \cos(\gamma - \alpha) + \sin(\alpha - \beta) \sin(\gamma - \alpha) \\ &= \left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \\ &= \frac{1}{4} + \frac{3}{4} \\ &= 1 \end{aligned}$$

Similarly,

$$\cos(2\beta - \gamma - \alpha) = 1$$

$$\cos(2\gamma - \alpha - \beta) = 1$$

Thus, we find:

$$\Sigma \cos(2\alpha - \beta - \gamma) = 1 + 1 + 1 = 3$$

Question32

$\arg \left[\frac{(1+i\sqrt{3})(-\sqrt{3}-i)}{(1-i)(-i)} \right]$ is equal to

Options:

A. $\frac{5\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{2\pi}{3}$

D. $\frac{-\pi}{2}$

Answer: B

Solution:

We have, $\arg \left[\frac{(1+i\sqrt{3})(-\sqrt{3}-i)}{(1-i)(-i)} \right]$

$$\begin{aligned} \text{Let } z &= \frac{(1+i\sqrt{3})(-\sqrt{3}-i)}{(1-i)(-i)} \\ &= \frac{(i+i^2\sqrt{3})(-\sqrt{3}-i)}{(1-i)(-i^2)} \\ &= \frac{(-\sqrt{3}+i)(-\sqrt{3}-i)}{(1-i)} \\ &= \frac{(-\sqrt{3})^2 - i^2}{1-i} = \frac{3+1}{1-i} = \left(\frac{4}{1-i} \right) \\ &= \frac{4(1+i)}{(1-i)(1+i)} = 4 \times \frac{1+i}{2} \end{aligned}$$

$$\Rightarrow z = 2(1+i)$$

$$\begin{aligned} \therefore \arg(z) &= \arg(2+2i) = \tan^{-1}(1) \\ &= \tan^{-1}\left(\tan \frac{\pi}{4}\right) \end{aligned}$$

$$\Rightarrow \arg(z) = \frac{\pi}{4}$$

Question33

If $P(x, y)$ represents the complex number $z = x + iy$ in the argand plane and $\arg\left(\frac{z-3i}{z+4}\right) = \frac{\pi}{2}$, then the equation of the locus of P is

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Options:

A. $x^2 + y^2 + 4x - 3y = 0$ and $3x - 4y > 0$

B. $x^2 + y^2 + 4x - 3y + 2 = 0$ and $3x - 4y > 0$

C. $x^2 + y^2 + 4x - 3y = 0$ and $3x - 4y < 0$

D. $x^2 + y^2 + 4x - 3y + 2 = 0$ and $3x - 4y < 0$

Answer: C

Solution:

Given, $\arg\left(\frac{z-3i}{z+4}\right) = \frac{\pi}{2}$ and $z = x + iy$

$$\begin{aligned}\text{Now, let } \omega &= \frac{z-3i}{z+4} = \frac{x+iy-3i}{x+iy+4} \\&= \frac{x+(y-3)i}{(x+4)+iy} \times \frac{(x+4)-iy}{(x+4)-iy} \\&= \frac{x(x+4) - ixy + (y-3)(x+4)i + y(y-3)}{(x+4)^2 + y^2} \\&\Rightarrow \frac{x(x+4) + y(y-3)}{(x+4)^2 + y^2} + i \frac{(y-3)(x+4) - xy}{(x+4)^2 + y^2}\end{aligned}$$

Now,

$$\begin{aligned}\arg(\omega) &= \tan^{-1}\left(\frac{(y-3)(x+4) - xy}{x(x+4) + y(y-3)}\right) = \frac{\pi}{2} \\&\Rightarrow \frac{(y-3)(x+4) - xy}{x(x+4) + y(y-3)} = \tan \frac{\pi}{2} \\&\Rightarrow \frac{4y - 3x - 12}{x^2 + y^2 + 4x - 3y} = \infty\end{aligned}$$

So, clearly

$$x^2 + y^2 + 4x - 3y = 0$$

$$\text{and } 3x - 4y < 0$$

Question34

If $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are the roots of $x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$, then $\frac{1}{\alpha_1^2} + \frac{1}{\alpha_2^2} + \frac{1}{\alpha_3^2} + \frac{1}{\alpha_4^2} + \frac{1}{\alpha_5^2}$ is equal to

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Options:

A. 15

B. $\frac{1}{7}$

C. 7

D. 12

Answer: C

Solution:

Given,

$\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are roots of the equation

$$\begin{aligned}
& x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0 \\
\Rightarrow & x^5 - x^4 - 4x^4 + 4x^3 + 5x^3 - 5x^2 - 4x^2 + 4x + x - 1 = 0 \\
\Rightarrow & x^4(x-1) - 4x^3(x-1) + 5x^2(x-1) - 4x(x-1) + 1(x-1) = 0 \\
\Rightarrow & (x-1)[x^4 - 4x^3 + 5x^2 - 4x + 1] = 0 \\
\Rightarrow & (x-1)(x^2 - 3x + 1)(x^2 - x + 1) = 0 \\
\Rightarrow & x = 1, \frac{3 + \sqrt{5}}{2}, \frac{3 - \sqrt{5}}{2}, \frac{1 + \sqrt{3}i}{2}
\end{aligned}$$

$$\text{and } \frac{1 - \sqrt{3}i}{2}$$

$$\alpha_1 = 1 \Rightarrow \alpha_1^2 = 1 - \frac{1}{\alpha_1^2} = 1$$

$$\alpha_2 = \frac{3 + \sqrt{5}}{2} \Rightarrow \alpha_2^2 = \frac{(3 + \sqrt{5})^2}{4}$$

$$\Rightarrow \frac{1}{\alpha_2^2} = \frac{4}{(3 + \sqrt{5})^2}$$

$$\begin{aligned} \Rightarrow \alpha_3 &= \frac{3 - \sqrt{5}}{2} \Rightarrow \alpha_3^2 = \frac{(3 - \sqrt{5})^2}{4} \\ \Rightarrow \frac{1}{\alpha_3^2} &= \frac{4}{(3 - \sqrt{5})^2} \end{aligned}$$

$$\alpha_4 = \frac{1 + \sqrt{3}i}{2}$$

$$\Rightarrow \alpha_4^2 = \frac{(1 + \sqrt{3}i)^2}{4}$$

$$\Rightarrow \frac{1}{\alpha_4^2} = \frac{4}{(1 + \sqrt{3}i)^2}$$

$$\alpha_5 = \frac{1 - \sqrt{3}i}{2} \Rightarrow \alpha_5^2 = \frac{(1 - \sqrt{3}i)^2}{4}$$

$$\Rightarrow \frac{1}{\alpha_5^2} = \frac{4}{(1 - \sqrt{3}i)^2}$$

$$\begin{aligned}
\text{So, } & \frac{1}{\alpha_1^2} + \frac{1}{\alpha_2^2} + \frac{1}{\alpha_3^2} + \frac{1}{\alpha_4^2} + \frac{1}{\alpha_5^2} \\
&= 1 + \frac{4}{(3 + \sqrt{5})^2} + \frac{4}{(3 - \sqrt{5})^2} + \frac{4}{(1 + \sqrt{3}i)^2} + \frac{4}{(1 - \sqrt{3}i)^2} \\
&= 1 + 4 \left[\frac{3^2 + (\sqrt{5})^2 - 6\sqrt{5} + 3^2 + (\sqrt{5})^2 + 6\sqrt{5}}{(9 - 5)^2} + \frac{1 - 3 - 2\sqrt{3}i + 1 - 3 + 2\sqrt{3}i}{(1 + 3)^2} \right] \\
&= 1 + 4 \left[\frac{28}{16} + \frac{-4}{16} \right] \\
&= 1 + 4 \left(\frac{28 - 4}{16} \right) = 1 + 4 \times \frac{24}{16} = 7
\end{aligned}$$

Question 35

If Z is a complex number such that $|Z| \leq 3$ and $\frac{-\pi}{2} \leq \text{amp } Z \leq \frac{\pi}{2}$, then the area of the region formed by locus of Z is

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Options:

A. 9π

B. $\frac{9\pi}{2}$

C. 3π

D. $\frac{9\pi}{4}$

Answer: B

Solution:

$|Z| \leq 3 \Rightarrow$ Complex number Z lies within a circle centered at the origin with radius 3 .

$-\frac{\pi}{2} \leq \arg(z) \leq \frac{\pi}{2}$. This restricts z to the right half of the complex plane.

The intersection of these region is a semicircle with radius 3 . The area of semicircle is given by half the area of the full circle

The area of circle with radius r is $= \pi r^2$

$$= \pi(3)^2 = 9\pi$$

Half of this area $= \frac{9\pi}{2}$

So, area of region $= \frac{9\pi}{2}$ sq units.

Question36

The locus of the complex number Z such that $\arg\left(\frac{Z-1}{Z+1}\right) = \frac{\pi}{4}$ is

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Options:

A. a straight line

B. a circle

C. a parabola

D. an ellipse

Answer: B

Solution:

To determine the locus of the complex number Z , where $Z = x + iy$ (with x and y as real numbers), we analyze the given condition:

$$\arg\left(\frac{Z-1}{Z+1}\right) = \frac{\pi}{4}$$

This equation implies that the complex number $A = \frac{Z-1}{Z+1}$ forms an angle of $\frac{\pi}{4}$ with the positive real axis. Therefore, A lies on a line making an angle of $\frac{\pi}{4}$.

To simplify, we express:

$$\frac{Z-1}{Z+1} = e^{i\frac{\pi}{4}} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

Thus, we have:

$$\left|\frac{Z-1}{Z+1}\right| = \left|\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}}\right| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1$$

This calculation indicates that the complex number $\frac{Z-1}{Z+1}$ is on the unit circle. Therefore, we equate:

$$\left| \frac{Z-1}{Z+1} \right| = 1 \Rightarrow |Z-1| = |Z+1|$$

This describes the locus as the perpendicular bisector of the line segment joining 1 and -1, which corresponds to the imaginary axis $x = 0$. Thus, the locus is a straight line on the imaginary axis.

Question37

All the values of $(8i)^{\frac{1}{3}}$ are

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Options:

A. $\pm(\sqrt{3} + i), -2i$

B. $\pm\sqrt{3} + i, -2i$

C. $\pm(\sqrt{3} - i), 2i$

D. $\pm(2 + i), i$

Answer: B

Solution:

To find the values of $(8i)^{\frac{1}{3}}$, we utilize De Moivre's theorem. Express $8i$ in polar form as $8e^{i(\frac{\pi}{2} + 2k\pi)}$, then take the cube root:

$$(8i)^{\frac{1}{3}} = 8^{\frac{1}{3}} e^{i(\frac{\theta + 2k\pi}{3})} = 2e^{i(\frac{\pi}{6} + \frac{2k\pi}{3})}$$

Next, calculate this for different values of k :

For $k = 0$:

$$2e^{i(\frac{\pi}{6})} = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = \sqrt{3} + i$$

For $k = 1$:

$$2e^{i(\frac{\pi}{6} + \frac{2\pi}{3})} = 2e^{i(\frac{5\pi}{6})} = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -\sqrt{3} + i$$

For $k = 2$:

$$2e^{i(\frac{\pi}{6} + \frac{4\pi}{3})} = 2e^{i(\frac{3\pi}{2})} = 2(0 - i) = -2i$$

Therefore, all the values of $(8i)^{\frac{1}{3}}$ are $\sqrt{3} + i$, $-\sqrt{3} + i$, and $-2i$, or in other words, $\pm\sqrt{3} + i, -2i$.

Question38

If the number of real roots of $x^9 - x^5 + x^4 - 1 = 0$ is n , the number of complex roots having argument on imaginary axis is m and the number of complex roots having argument in 2nd quadrant is K , $m \cdot n \cdot k =$

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Options:

- A. 6
- B. 9
- C. 12
- D. 24

Answer: A

Solution:

$$x^9 - x^5 + x^4 - 1 = 0$$

$$\Rightarrow x^5(x^4 - 1) + 1(x^4 - 1) = 0$$

$$\Rightarrow (x^5 + 1)(x^4 - 1) = 0$$

$$\Rightarrow x^5 + 1 = 0 \text{ or } x^4 - 1 = 0$$

$$\Rightarrow x^5 = -1 \text{ or } x^4 = 1$$

$$\Rightarrow x^5 = -1 \text{ or } x = 1, -1, i, -i$$

$\therefore$ Number of real roots (n) = 3

Number of complex roots (m) = 2

Number of complex roots having argument in 2nd quadrant (k) = 1

$$\therefore m \cdot n \cdot k = 3(2)(1) = 6$$

Question39

Imaginary part of $\frac{(1-i)^3}{(2-i)(3-2i)}$ is

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Options:

- A. $\frac{22}{65}$
- B. $\frac{6}{65}$
- C. $-\frac{6}{65}$
- D. $-\frac{22}{65}$

Answer: D

Solution:

We have,

$$\begin{aligned}
\frac{(1-i)^3}{(2-i)(3-2i)} &= \frac{1^3 - i^3 - 3 \cdot 1 \cdot i(1-i)}{(6-4i-3i+2i^2)} \\
&= \frac{[\because (a-b)^3 = a^3 - b^3 - 3ab(a-b)]}{(6-7i-2)} = \frac{1+i-3i-3}{4-7i} \\
&= \frac{-2-2i}{4-7i} \\
&= \frac{-2-2i}{4-7i} \times \frac{4+7i}{4+7i} \quad [rationalising] \\
&= \frac{-(2+2i)(4+7i)}{4^2 - (7i)^2} \\
&= \frac{-(8+14i+8i-14)}{16+49} = \frac{6-22i}{65}
\end{aligned}$$

On comparing with $Z = x + iy$, we get

Imaginary part is $\frac{-22}{65}$.

Question 40

The square root of $7 + 24i$

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Options:

A. $4 - 3i$

B. $3 + 4i$

C. $3 - 4i$

D. $4 + 3i$

Answer: D

Solution:

To find the square root of $7 + 24i$, set it equal to $x + yi$, where $x, y \in \mathbb{R}$.

Squaring both sides gives:

$$(x + yi)^2 = x^2 - y^2 + 2xyi$$

This leads to:

$$x^2 - y^2 + 2xyi = 7 + 24i$$

By equating the real and imaginary parts, we obtain:

$$x^2 - y^2 = 7 \quad (\text{equation 1})$$

$$2xy = 24 \quad \Rightarrow \quad xy = 12 \quad (\text{equation 2})$$

From equation 2, solve for y :

$$y = \frac{12}{x}$$

Substitute $y = \frac{12}{x}$ into equation 1:

$$x^2 - \left(\frac{12}{x}\right)^2 = 7$$

Simplifying the equation, we get:

$$x^2 - \frac{144}{x^2} = 7$$

$$x^4 - 144 = 7x^2$$

$$x^4 - 7x^2 - 144 = 0$$

This can be factored as:

$$(x^2 - 16)(x^2 + 9) = 0$$

Since x is real, we discard $x^2 + 9 = 0$ as it does not lead to real solutions for x . Therefore:

$$x^2 = 16 \Rightarrow x = \pm 4$$

For $x = 4, y = 3$.

For $x = -4, y = -3$.

Thus, the square roots of $7 + 24i$ are $\pm(4 + 3i)$.

Therefore, the correct solution is $4 + 3i$.

Question41

If n is an integer and $Z = \cos \theta + i \sin \theta, \theta \neq (2n + 1) \frac{\pi}{2}$, then $\frac{1+Z^{2n}}{1-Z^{2n}} =$

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Options:

A. $i \tan n\theta$

B. $i \cot n\theta$

C. $-i \tan n\theta$

D. $-i \cot n\theta$

Answer: D

Solution:

We have,

$$Z = \cos \theta + i \sin \theta, \theta \neq (2n + 1) \frac{\pi}{2}$$

$$\text{Now, } \frac{1+Z^{2n}}{1-Z^{2n}} = \frac{1+(\cos \theta + i \sin \theta)^{2n}}{1-(\cos \theta + i \sin \theta)^{2n}}$$

$$= \frac{1+(\cos 2n\theta + i \sin 2n\theta)}{1-(\cos 2n\theta + i \sin 2n\theta)}$$

[By De Moivre's theorem]

$$\begin{aligned}
&= \frac{1 + (2 \cos^2 n\theta - 1) + 2i \sin n\theta \cos n\theta}{1 - (1 - 2 \sin^2 n\theta) - 2i \sin n\theta \cos n\theta} \\
&\quad [\cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A] \\
&= \frac{2 \cos n\theta [\cos n\theta + i \sin n\theta]}{2 \sin n\theta [\sin n\theta - i \cos n\theta]} \\
&= -i \cot n\theta \left[\frac{\cos n\theta + i \sin n\theta}{\cos n\theta + i \sin n\theta} \right] \\
&= -i \cot n\theta.
\end{aligned}$$

Question42

The complex conjugate of $(4 - 3i)(2 + 3i)(1 + 4i)$ is.

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Options:

- A. $7 + 74i$
- B. $-7 + 74i$
- C. $-7 - 74i$
- D. $7 - 74i$

Answer: C

Solution:

When calculating the complex conjugate of the expression $(4 - 3i)(2 + 3i)(1 + 4i)$, follow these steps:

First, compute the product:

$$(4 - 3i)(2 + 3i) = 8 + 12i - 6i - 9i^2 = 8 + 6i + 9 = 17 + 6i$$

Here, we used the fact that $i^2 = -1$.

Next, multiply this result by $(1 + 4i)$:

$$(17 + 6i)(1 + 4i) = 17 + 68i + 6i + 24i^2 = 17 + 74i - 24$$

Simplifying this, we get:

$$-7 + 74i$$

To find the complex conjugate of $-7 + 74i$, change the sign of the imaginary part:

$$-7 - 74i$$

So, the complex conjugate of the expression is $-7 - 74i$.

Question43

If the amplitude of $(z - 2)$ is $\frac{\pi}{2}$, then the locus of z is

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Options:

A. $x = 0, y > 0$

B. $x = 2, y > 0$

C. $x > 0, y = 2$

D. $x > 0, y = 0$

Answer: B

Solution:

To determine the locus of z , given that the amplitude (also known as the argument) of $(z - 2)$ is $\frac{\pi}{2}$, we proceed as follows:

Let $z = x + iy$, where x and y are real numbers. Given the condition:

$$\arg(z - 2) = \frac{\pi}{2}$$

Substituting $z = x + iy$ into the expression, we have:

$$\arg((x - 2) + iy) = \frac{\pi}{2}$$

This implies:

$$\tan^{-1}\left(\frac{y}{x-2}\right) = \frac{\pi}{2}$$

From the property of arctangent, we know:

$$\frac{y}{x-2} \rightarrow \infty$$

This scenario occurs when $x - 2 = 0$. Hence,

$$x = 2$$

Thus, the locus of z is a vertical line at $x = 2$, which is parallel to the Y-axis, only considered for $y > 0$.

Therefore, the locus of z is a vertical line $x = 2$ for $y > 0$.

Question44

If ω is the cube root of unity, $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} =$

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Options:

A. 2

B. -2

C. 1

D. -1

Answer: D

Solution:

If ω is the cube root of unity.

$$\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} = ?$$

Now, we write as

$$A = a + b\omega + c\omega^2$$

$$B = b + c\omega + a\omega^2$$

$$C = c + a\omega + b\omega^2$$

Find common denominator for the fractions.

$$\frac{A}{C} + \frac{A}{B} = \frac{A \cdot B + A \cdot C}{B \cdot C} = \frac{A(B+C)}{BC}$$

Now, we know that according to given condition.

$$\omega^3 = 1$$

$$1 + \omega + \omega^2 = 0$$

$$B + C = (b + c\omega + a\omega^2) + (c + a\omega + b\omega^2)$$

$$b + c\omega + a\omega^2 + c + a\omega + b\omega^2$$

$$= (b + \omega^2 b) + (c\omega + c) + (a\omega^2 + a\omega)$$

$$= b(1 + \omega^2) + c(1 + \omega) + a(\omega^2 + \omega)$$

$$(\because \text{using } 1 + \omega + \omega^2 = 0)$$

$$1 + \omega^2 = -\omega, 1 + \omega = -\omega^2$$

$$B + C = b(-\omega) + c(-\omega^2) + a(-1)$$

$$= -a - b\omega - c\omega^2$$

$$\text{Thus, } B + C = -A$$

$$\text{Because, } A = a + b\omega + c\omega^2$$

Substituting this into the expression

$$\frac{A(B+C)}{BC} = \frac{A(-A)}{BC} = \frac{-A^2}{BC}$$

The denominator (BC) does not change the overall simplification that each term involves $\bar{A}$, B and C with symmetrical properties.

$$\text{Thus, } \frac{A}{C} + \frac{A}{B} = -1$$

Question45

If $(3 + i)$ is a root of $x^2 + ax + b = 0$, then $a =$

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Options:

A. 3

B. -3

C. 6

D. -6

Answer: D

Solution:

$$\begin{aligned}
&\text{Given, } (3+i) \text{ is a root of} \\
&x^2 + ax + b = 0 \\
&(3+i)^2 + a(3+i) + b = 0 \\
&9 - 1 + 6i + a(3+i) + b = 0 \\
&8 + 6i + 3a + ai + b = 0 \\
&(8 + 3a + b) + i(6 + a) = 0 \\
&3a + b = -8 \\
&a + 6 = 0 \Rightarrow a = -6
\end{aligned}$$

Question 46

If $z_1 = 10 + 6i$, $z_2 = 4 + 6i$ and z is any complex number such that the argument of $\frac{(z-z_1)}{(z-z_2)}$ is $\frac{\pi}{4}$,

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Options:

A. $|z - 7 - 9i| = 3\sqrt{2}$

B. $|z - 7 - 9i| = 2\sqrt{2}$

C. $|z - 3 + 9i| = 3\sqrt{2}$

D. $|z + 3 - 9i| = 2\sqrt{2}$

Answer: A

Solution:

Given, $z_1 = 10 + 6i$, $z_2 = 4 + 6i$

and $\arg\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$

Let $z = x + iy$

$$\begin{aligned}
\text{Now, } \frac{z-z_1}{z-z_2} &= \frac{(x+iy)-(10+6i)}{(x+iy)-(4+6i)} \\
&= \frac{(x-10) + i(y-6)}{(x-4) + i(y-6)} \\
&= \frac{(x-10) + i(y-6)}{(x-4) + i(y-6)} \times \frac{(x-4) - i(y-6)}{(x-4) - i(y-6)} \\
&= \frac{(x-10)(x-4) + (y-6)^2 + i[-(x-10)(y-6) + (y-6)(x-4)]}{(x-4)^2 + (y-6)^2}
\end{aligned}$$

Since, $\arg\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$

i.e. $\tan \theta = 1$

$$\therefore \frac{(y-6)(x-4) - (x-10)(y-6)}{(x-10)(x-4) + (y-6)^2} = 1$$

$$\Rightarrow \frac{xy - 6x - 4y + 24 - xy + 6x + 10y - 60}{x^2 - 14x + 40 + y^2 - 12y + 36} = 1$$

$$\Rightarrow 6y - 36 = x^2 + y^2 - 12y - 14x + 76$$

$$\Rightarrow x^2 + y^2 - 18y - 14x + 112 = 0$$

$$\Rightarrow (x-7)^2 + (y-9)^2 = 18$$

$$\Rightarrow |z - (7 + 9i)| = 3\sqrt{2}$$

Question47

If $\frac{3-2i \sin \theta}{1+2i \sin \theta}$ is purely imaginary number, then $\theta =$

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Options:

A. $2n\pi \pm \frac{\pi}{4}$

B. $2n\pi \pm \frac{\pi}{2}$

C. $n\pi \pm \frac{\pi}{3}$

D. $n\pi \pm \frac{\pi}{6}$

Answer: C

Solution:

$$\frac{3-2i \sin \theta}{1+2i \sin \theta} = \frac{(3-2i \sin \theta)}{(1+2i \sin \theta)} \times \frac{(1-2i \sin \theta)}{(1-2i \sin \theta)}$$

(by rationalising)

$$\begin{aligned} &= \frac{3 - 6i \sin \theta - 2i \sin \theta - 4 \sin^2 \theta}{1 + 4 \sin^2 \theta} \\ &= \frac{3 - 4 \sin^2 \theta - 8i \sin \theta}{1 + 4 \sin^2 \theta} \end{aligned}$$

Since, $\frac{3-2i \sin \theta}{1+2i \sin \theta}$ is purely imaginary.

$$\therefore \frac{3-4 \sin^2 \theta}{1+4 \sin^2 \theta} = 0$$

$$\Rightarrow 3 - 4 \sin^2 \theta = 0 \Rightarrow 4 \sin^2 \theta = 3$$

$$\Rightarrow \sin^2 \theta = \frac{3}{4} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}$$

Question48

If $z = x + iy$, $x^2 + y^2 = 1$ and $z_1 = ze^{i\theta}$, then $\frac{z_1^{2n}-1}{z_1^{2n}+1} =$

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Options:

A. $-i \tan \left(n \left(\theta + \tan^{-1} \left(\frac{y}{x} \right) \right) \right)$

B. $i \cot \left(n \left(\theta + \tan^{-1} \left(\frac{y}{x} \right) \right) \right)$

C. $i \tan \left(n \left(\theta + \tan^{-1} \frac{x}{u} \right) \right)$

D. $i \tan \left(n \left(\theta + \tan^{-1} \frac{y}{x} \right) \right)$

Answer: D

Solution:

Given, $z = x + iy$, $x^2 + y^2 = 1$ and

$$z_1 = ze^{i\theta}$$

$$z_1 = (x + iy)e^{i\theta}$$

Taking $(2n)$ th power of both sides of the equation

$$\begin{aligned} z_1^{2n} &= (x + iy)^{2n} e^{i2n\theta} \\ &= r^{2n} (\cos(2n\theta) + i \sin(2n\theta)) \\ z_1^{2n} + 1 &= r^{2n} (\cos(2n\theta) + i \sin(2n\theta)) + 1 \\ &= (\cos(2n\theta) + i \sin(2n\theta)) + 1 \\ &[\because x^2 + y^2 = 1] \\ z_1^{2n} - 1 &= (\cos(2n\theta) + i \sin(2n\theta)) - 1 \\ \frac{z_1^{2n} - 1}{z_1^{2n} + 1} &= \frac{(\cos(2n\theta) + i \sin(2n\theta)) - 1}{(\cos(2n\theta) + i \sin(2n\theta)) + 1} \\ &= \frac{1 - 2\sin^2(n\theta) + i \sin(2n\theta) - 1}{2\cos^2(n\theta) - 1 + i \sin(2n\theta) + 1} \\ &= \frac{-2\sin^2(n\theta) + 2i \sin(n\theta) \cos(n\theta)}{2\cos^2(n\theta) + 2i \sin(n\theta) \cos(n\theta)} \\ &= \frac{[i \sin(n\theta) + \cos(n\theta)] i \sin(n\theta)}{[\cos(n\theta) + i \sin(n\theta)] \cos(n\theta)} \\ &= i \tan n\theta = i \tan \left(n \left(\theta + \tan^{-1} \frac{y}{x} \right) \right) \end{aligned}$$

Question 49

If the point P represents the complex number $z = x + iy$ in the argand plane and if $\frac{z+i}{z-i}$ is a purely imaginary number, then the locus of P is

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Options:

A. $x^2 + y^2 + x - y = 0$ and $(x, y) \neq (1, 0)$

B. $x^2 + y^2 - x + y = 0$ and $(x, y) \neq (1, 0)$

C. $x^2 + y^2 - x + y = 0$ and $(x, y) = (1, 0)$

D. $x^2 + y^2 + x + y = 0$

Answer: B

Solution:

We have, $\frac{z+i}{z-1} = \frac{x+iy+i}{x+iy-1}$

$$\begin{aligned}
&= \frac{x + i(1 + y)}{(x - 1) + iy} \times \frac{(x - 1) - iy}{(x - 1) - iy} \\
&= \frac{x(x - 1) + i(1 + y)(x - 1) - ixy + y(1 + y)}{(x - 1)^2 + y^2} \\
&= \frac{x^2 - x + y^2 + i(x + xy - y - 1 - xy)}{(x - 1)^2 + y^2} \\
&= \frac{x^2 + y^2 - x + y}{(x - 1)^2 + y^2} + i \frac{(x - y - 1)}{(x - 1)^2 + y^2}
\end{aligned}$$

Since, $\frac{z+i}{z-1}$ is purely imaginary.

$$\text{So, } \frac{x^2 + y^2 - x + y}{(x - 1)^2 + y^2} = 0$$

$$x^2 + y^2 - x + y = 0$$

Where, $x \neq 1, y \neq 0$

Question 50

$S = \{z \in \mathbb{C} / |z + 1 - i| = 1\}$ represents

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Options:

- A. the circle with centre at $(-1, 1)$ and radius 1 unit
- B. the circle with centre at $(1, -1)$ and radius 1 unit
- C. the closed circular disc with centre at $(1, -1)$ and radius 1 unit
- D. the closed circular disc with centre at $(-1, 1)$ and radius 1 unit

Answer: A

Solution:

The given set $S = \{z \in \mathbb{C} \mid |z + 1 - i| = 1\}$ can be analyzed as follows:

First, consider the expression $|z + 1 - i| = 1$.

Break down z in terms of real and imaginary components:

$$z = x + iy,$$

where x and y are real numbers. Thus, we can substitute z in the expression:

$$|x + iy + 1 - i| = 1$$

Simplifying inside the absolute value, we have:

$$|(x + 1) + i(y - 1)| = 1$$

The expression $|(x + 1) + i(y - 1)|$ represents the distance from the point $(-1, 1)$ in the complex plane. Therefore, the set of all such points for which this distance is 1 forms a circle centered at $(-1, 1)$ with a radius of 1. Mathematically, this can be simplified to:

$$\sqrt{(x + 1)^2 + (y - 1)^2} = 1$$

Squaring both sides, we get:

$$(x + 1)^2 + (y - 1)^2 = 1$$

Thus, S represents a circle with center at $(-1, 1)$ and radius 1 unit.

Question 51

If m, n are respectively the least positive and greatest negative integer value of k such that $\left(\frac{1-i}{1+i}\right)^k = -i$, then $m - n =$

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Options:

A. 4

B. 0

C. 6

D. 2

Answer: A

Solution:

To solve for the least positive and greatest negative integer values of k in the expression $\left(\frac{1-i}{1+i}\right)^k = -i$, follow these steps:

Start with the given equation:

$$\left(\frac{1-i}{1+i}\right)^k = -i$$

Simplify the fraction $\frac{1-i}{1+i}$:

$$\frac{1-i}{1+i} \cdot \frac{1-i}{1-i} = \frac{(1-i)^2}{(1+i)(1-i)}$$

Calculate the numerator and the denominator:

$$(1-i)^2 = 1 - 2i + i^2 = 1 - 2i - 1 = -2i$$

$$(1+i)(1-i) = 1 + i - i - i^2 = 1 + 1 = 2$$

Substitute back into the fraction:

$$\frac{(1-i)^2}{(1+i)(1-i)} = \frac{-2i}{2} = -i$$

Therefore, the expression simplifies to:

$$(-i)^k = -i$$

For $(-i)^k = -i$, it implies:

$$k \equiv 1 \pmod{4}$$

From this, determine the least positive integer m and greatest negative integer n :

$m = 1$ (since $k = 1$ is the smallest positive integer solution)

$n = -3$ (since $k = -3$ satisfies the condition $-i$ in the sequence and is the largest negative solution)

Calculate $m - n$:

$$m - n = 1 - (-3) = 4$$

Thus, $m - n = 4$.

Question 52

If a complex number z is such that $\frac{z-2i}{z-2}$ is purely imaginary number and the locus of z is a closed curve, then the area of the region bounded by that closed curve and lying in the first quadrant is $\frac{z-2i}{z-2}$

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Options:

A. 2π

B. $\frac{\pi}{2}$

C. π

D. $\frac{\pi}{4}$

Answer: A

Solution:

$\frac{z-2i}{z-2}$ is purely imaginary

Let $z = x + iy$

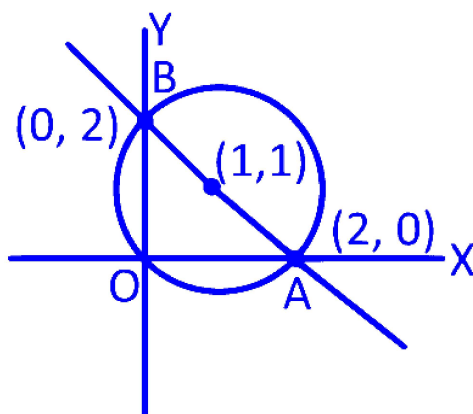
$$\therefore \frac{x+iy-2i}{x+iy-2} = \frac{x+i(y-2)}{(x-2)+iy} \times \frac{(x-2)-iy}{(x-2)-iy}$$

Real part of $\frac{z-2i}{z-2} = 0$

$$x(x-2) + y(y-2) = 0$$

$$\Rightarrow x^2 + y^2 - 2x - 2y = 0$$

$$\Rightarrow (x-1)^2 + (y-1)^2 = 2, \text{ which is equation of circle with radius } \sqrt{2} \text{ units.}$$



$$\text{Required area} = \frac{1}{2} \times 2 \times 2 + \frac{\pi}{2} (\sqrt{2})^2 = (2 + \pi) \text{ sq units}$$

Question53

Real part of $\frac{(\cos a + i \sin a)^6}{(\sin b + i \cos b)^8}$ is

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Options:

A. $\sin(6a - 8b)$

B. $\cos(6a - 8b)$

C. $\sin(6a + 8b)$

D. $\cos(6a + 8b)$

Answer: D

Solution:

$$\begin{aligned} \text{We have, } & \frac{(\cos a + i \sin a)^6}{(\sin b + i \cos b)^8} \\ &= \frac{(\cos a + i \sin a)^6}{(-i^2 \sin b + i \cos b)^8} \quad [\because 1 = -i^2] \\ &= \frac{(\cos a + i \sin a)^6}{i^8 (\cos b - i \sin b)^8} \\ &= \frac{(\cos a + i \sin a)^6}{(\cos b - i \sin b)^8} \\ &= \frac{\cos 6a + i \sin 6a}{\cos 8b - i \sin 8b} \\ &= \frac{(\cos 6a + i \sin 6a)(\cos 8b + i \sin 8b)}{(\cos 8b - i \sin 8b)(\cos 8b + i \sin 8b)} \\ \therefore \text{ Required real part} &= \frac{\cos 6a \cos 8b - \sin^8 6a \sin 8b}{\cos^2 8b + \sin^2 8b} \\ &= \cos(6a + 8b) \end{aligned}$$

Question54

If real parts of $\sqrt{-5 - 12i}$, $\sqrt{5 + 12i}$ are positive values, the real part of $\sqrt{-8 - 6i}$ is a negative value and $a + ib = \frac{\sqrt{-5-12i} + \sqrt{5+12i}}{\sqrt{-8-6i}}$, then $2a + b =$

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Options:

A. 3

B. 2

C. -3

D. -2

Answer: C

Solution:

To find $a + ib = \frac{\sqrt{-5-12i} + \sqrt{5+12i}}{\sqrt{-8-6i}}$ and calculate $2a + b$:

Calculate $\sqrt{-5-12i}$ and $\sqrt{5+12i}$:

Since the real parts of $\sqrt{-5-12i}$ and $\sqrt{5+12i}$ are positive:

$$\sqrt{-5-12i} = \sqrt{\frac{13-5}{2}} + i\sqrt{\frac{13+5}{2}} = 2 - 3i$$

$$\sqrt{5+12i} = \sqrt{\frac{13+5}{2}} + i\sqrt{\frac{13-5}{2}} = 3 + 2i$$

Calculate $\sqrt{-8-6i}$:

Given that the real part is negative:

$$\sqrt{-8-6i} = -\sqrt{\frac{10-8}{2}} + i\sqrt{\frac{10+8}{2}} = -1 + 3i$$

Find $a + ib$:

Using the calculations from the previous steps:

$$a + ib = \frac{(2-3i)+(3+2i)}{-1+3i} = \frac{5-i}{-1+3i}$$

Multiply the numerator and the denominator by the conjugate of the denominator:

$$= \frac{(5-i)(-1-3i)}{(-1+3i)(-1-3i)}$$

Simplify:

$$= \frac{-5+i-15i-3}{1+9} = \frac{-8-14i}{10}$$

$$= -\frac{4}{5} - \frac{7i}{5}$$

Thus, $a = -\frac{4}{5}$ and $b = -\frac{7}{5}$.

Calculate $2a + b$:

$$2a + b = 2\left(-\frac{4}{5}\right) + \left(-\frac{7}{5}\right) = -\frac{8}{5} - \frac{7}{5} = -3$$

Hence, the result is -3 .

Question55

The set of all real values of c for which the equation $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + c = 0$ represents a circle, is

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Options:

A. $[25, 50]$

B. $[-5, 5]$

C. $[-20, -5] \cup [5, 20]$

D. $[-25]$

Answer: D

Solution:

The given equation $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + c = 0$ is analyzed to determine the real values of c that form a circle.

Key Steps

Identifying the Circle's Center:

$z\bar{z}$ is the modulus squared of z , and the terms $(4 - 3i)z$ and $(4 + 3i)\bar{z}$ suggest the transformation involves the complex number $4 - 3i$.

Thus, the center of the circle is at $4 - 3i$.

Finding the Radius:

To find the radius, use the formula that incorporates the modulus of the center:

$$\sqrt{(4)^2 + (-3)^2 - c} = \sqrt{25 - c}$$

The radius needs to be non-negative for the equation to represent a circle:

$$\sqrt{25 - c} \geq 0$$

which simplifies to:

$$25 - c \geq 0 \quad \Rightarrow \quad c \leq 25$$

Conclusion

The real values of c for which the equation represents a circle are in the interval:

$$(-\infty, 25]$$

Question 56

If $z = x + iy$ is a complex number, then the number of distinct solutions of the equation $z^3 + \bar{z} = 0$ is

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Options:

A. 1

B. 3

C. Infinite

D. 5

Answer: D

Solution:

Given the equation $z^3 + \bar{z} = 0$, where $z = x + iy$ is a complex number, we can find the distinct solutions as follows:

Rewrite the Equation:

Start with:

$$z^3 + \bar{z} = 0$$

This implies:

$$z^3 = -\bar{z}$$

Equate Magnitudes:

Taking the magnitude of both sides, we have:

$$|z^3| = |-\bar{z}|$$

This simplifies to:

$$|z|^3 = |\bar{z}|$$

Since $|\bar{z}| = |z|$, the equation becomes:

$$|z|^3 = |z|$$

Solve for Magnitude:

This equation leads to:

$$|z|(|z|^2 - 1) = 0$$

Therefore, either $|z| = 0$ or $|z|^2 = 1$.

Consider Each Case:

If $|z| = 0$, then $z = 0$. This gives one distinct solution: $z = 0$.

If $|z|^2 = 1$, then $|z| = 1$ and consequently $z\bar{z} = 1$. This implies:

$$z = \frac{1}{\bar{z}}$$

Substitute Back:

Substituting $\bar{z} = \frac{1}{z}$ into the original equation:

$$z^3 + \frac{1}{z} = 0$$

Multiply through by z to clear the fraction:

$$z^4 + 1 = 0$$

Solve $z^4 + 1 = 0$:

This equation has four roots:

$$z = e^{i(\pi/4)}, e^{i(3\pi/4)}, e^{i(5\pi/4)}, e^{i(7\pi/4)}$$

Count Distinct Solutions:

Including $z = 0$, we have:

One solution for $z = 0$

Four solutions from $z^4 + 1 = 0$

Therefore, the number of distinct solutions to the equation is 5.

Question 57

By simplifying $i^{18} - 3i^7 + i^2(1 + i^4)(i)^{22}$, we get

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Options:

A. $-1 + 3i$

B. $1 - 3i$

C. $1 + 3i$

D. $-1 - 3i$

Answer: C

Solution:

$$\begin{aligned} & i^{18} - 3i^7 + i^2(1 + i^4)(i)^{22} \\ &= i^{4 \times 4 + 2} - 3i^{4 + 3} + -1(1 + 1)(i^2)^{11} \\ &= i^2 - 3i^3 + 2 \quad [\because i^2 = -1 \text{ and } i^4 = 1] \\ &= -1 + 3i + 2 = 1 + 3i \end{aligned}$$

Question 58

The values of x for which $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other are

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Options:

A. $x = n\pi \pm \frac{\pi}{6}$

B. None

C. $x = n\pi \pm \frac{\pi}{3}$

D. $x = \left(n + \frac{1}{2}\right)\pi$

Answer: B

Solution:

Conjugate of $\cos x - i \sin 2x$ is $\cos x + i \sin 2x$

So, $\sin x + i \cos 2x = \cos x + i \sin 2x$

Comparing real and imaginary parts,

$$\sin x = \cos x \text{ or } \tan x = 1$$

$$\text{and } \cos 2x = \sin 2x \text{ or } \tan 2x = 1$$

But for same value of x , both $\tan x$ and $\tan 2x$ can not. Hence, no solution is possible.

Question 59

The locus of a point z satisfying $|z|^2 = \operatorname{Re}(z)$ is a circle with centre

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Options:

- A. $(0, \frac{1}{2})$
- B. $(-\frac{1}{2}, 0)$
- C. $(\frac{1}{2}, 0)$
- D. $(0, -\frac{1}{2})$

Answer: C

Solution:

$$\text{Let } z = x + iy$$
$$|z| = \sqrt{x^2 + y^2}$$

$$\text{Now, } |z|^2 = \operatorname{Re}(z)$$

$$x^2 + y^2 = x$$
$$x^2 + y^2 - x = 0$$
$$g = 1/2, f = 0$$

So, centre of circle $(\frac{1}{2}, 0)$.

Question 60

Multiplicative inverse of the complex number $(\sin \theta, \cos \theta)$ is

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Options:

- A. $(\sin \theta, \cos \theta)$
- B. $(\sin \theta, -\cos \theta)$
- C. $(\cos \theta, -\sin \theta)$
- D. $(-\cos \theta, \sin \theta)$

Answer: B

Solution:

Multiplicative inverse of complex number $(\sin \theta, \cos \theta)$ is

$$\begin{aligned} z &= \sin \theta + i \cos \theta \\ \Rightarrow \frac{1}{z} &= \frac{(\sin \theta - i \cos \theta)}{(\sin \theta + i \cos \theta)(\sin \theta - i \cos \theta)} \\ &= \frac{(\sin \theta - i \cos \theta)}{\sin^2 \theta + \cos^2 \theta} = (\sin \theta - i \cos \theta) \end{aligned}$$

Thus, multiplicative inverse of $(\sin \theta, \cos \theta)$ is $(\sin \theta, -\cos \theta)$.

Question61

$$\sum_{k=0}^{440} i^k = x + iy \Rightarrow x^{100} + x^{99}y + x^{242}y^2 + x^{97}y^3 =$$

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Options:

- A. 0
- B. -4
- C. 4
- D. 1

Answer: D

Solution:

$$\begin{aligned} \sum_{k=0}^{440} i^k &= x + iy \\ \Rightarrow x + iy &= 1 + i + i^2 + i^3 + i^4 + \dots + i^{440} = -1 \\ \therefore x &= -1, y = 0 \\ \text{Then, } x^{100} + x^{99} \cdot y + x^{242} \cdot y^2 + x^{97} \cdot y^3 & \\ \Rightarrow (-1)^{100} + 0 &= 1 \end{aligned}$$

Question62

If $e^{i\theta} = \text{cis } \theta$, then $\sum_{n=0}^{\infty} \frac{\cos(n\theta)}{2^n} =$

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Options:

A. $(4 + 2 \cos \theta)/(5 - 4 \cos \theta)$

B. $(4 - 2 \cos \theta)/(5 + 4 \cos \theta)$

C. $(4 - 2 \cos \theta)/(5 - 4 \cos \theta)$

D. $(4 + 2 \cos \theta)/(5 + 4 \cos \theta)$

Answer: C

Solution:

$$e^{i\theta} = \operatorname{cis} \theta = \cos \theta + i \sin \theta$$

$$\sum_{n=0}^{\infty} \frac{\cos(n\theta)}{2^n}$$

$$= \operatorname{Re} \left[\sum_{n=0}^{\infty} \frac{\cos n\theta}{2^n} \right]$$

$$= \operatorname{Re} \left[\sum_{n=0}^{\infty} \left(\frac{\cos \theta}{2} \right)^n \right] \quad [\text{Using De-Moivre theorem}]$$

$$= \operatorname{Re} \left[\sum_{n=0}^{\infty} \left(\frac{e^{i\theta}}{2} \right)^n \right]$$

$$= \operatorname{Re} \left[1 + \frac{e^{i\theta}}{2} + \left(\frac{e^{i\theta}}{2} \right)^2 + \dots \infty \right]$$

$$1 + \frac{e^{i\theta}}{2} + \left(\frac{e^{i\theta}}{2} \right)^2 + \dots \text{ is an G P}$$

$$= \operatorname{Re} \left[\frac{1}{1 - \frac{e^{i\theta}}{2}} \right] = \operatorname{Re} \left[\frac{2}{2 - e^{i\theta}} \right]$$

$$= \operatorname{Re} \left[\frac{2}{2 - \cos \theta - i \sin \theta} \right]$$

$$= \operatorname{Re} \left[\frac{2(2 - \cos \theta - i \sin \theta)}{(2 - \cos \theta)^2 - i^2 \sin^2 \theta} \right]$$

$$= \operatorname{Re} \left[\frac{4 - 2 \cos \theta}{4 + \cos^2 \theta - 4 \cos \theta + \sin^2 \theta} \right]$$

$$+ \frac{i \sin \theta}{4 + \cos^2 \theta - 4 \cos \theta + \sin^2 \theta} \right]$$

$$= \operatorname{Re} \left[\frac{4 - 2 \cos \theta}{5 - 4 \cos \theta} + \frac{i \sin \theta}{5 - 4 \cos \theta} \right]$$

$$= \frac{4 - 2 \cos \theta}{5 - 4 \cos \theta}$$

Question 63

$$iz^3 + z^2 - z + i = 0 \Rightarrow |z| =$$

Options:

- A. $1/2$
- B. 2
- C. $3/2$
- D. 1

Answer: D

Solution:

$$\begin{aligned} & iz^3 + z^2 - z + i = 0 \\ \Rightarrow & iz^3 - z + z^2 + i = 0 \\ \Rightarrow & iz^3 + i^2z + z^2 + i = 0 \\ \Rightarrow & iz(z^2 + i) + (z^2 + i) = 0 \\ \Rightarrow & (z^2 + i)(iz + 1) = 0 \\ \Rightarrow & (z^2 + i)(iz - i^2) = 0 \\ \Rightarrow & (z^2 + i)i(z - i) = 0 \\ \Rightarrow & z^2 = -i \text{ or } z = i \\ & z^2 = 1 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) \\ \text{or } & z = 1 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \end{aligned}$$

We can see that all the roots have modulus as 1.

So, $|z| = 1$

Question64

If $\frac{x-1}{3+i} + \frac{y-1}{3-i} = i$, then the true statement among the following is

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Options:

- A. $x < 0, y < 0$
- B. $x < 0, y > 0$
- C. $x > 0, y < 0$
- D. $x > 0, y > 0$

Answer: B

Solution:

Given

$$\frac{x-1}{3+i} + \frac{y-1}{3-i} = i \quad (x, y \text{ are real})$$

Step 1: Rationalize the denominators (find reciprocals)

$$\frac{1}{3+i} = \frac{3-i}{(3+i)(3-i)} = \frac{3-i}{10}$$
$$\frac{1}{3-i} = \frac{3+i}{(3-i)(3+i)} = \frac{3+i}{10}$$

So the equation becomes:

$$(x-1)\frac{3-i}{10} + (y-1)\frac{3+i}{10} = i$$

Multiply by 10:

$$(x-1)(3-i) + (y-1)(3+i) = 10i$$

Step 2: Expand and separate Real & Imaginary parts

$$(x-1)(3-i) = 3(x-1) - i(x-1)$$

$$(y-1)(3+i) = 3(y-1) + i(y-1)$$

Add them:

$$3(x-1) + 3(y-1) + i(-(x-1) + (y-1)) = 10i$$

✔ Real part:

$$3(x-1) + 3(y-1) = 0 \Rightarrow 3(x+y-2) = 0 \Rightarrow x+y = 2$$

✔ Imaginary part:

$$-(x-1) + (y-1) = 10 \Rightarrow -x+1+y-1 = 10 \Rightarrow y-x = 10$$

Step 3: Solve the two equations

$$x+y = 2$$

$$y-x = 10$$

Add both:

$$2y = 12 \Rightarrow y = 6$$

Then

$$x = 2 - y = 2 - 6 = -4$$

Conclusion

$$x = -4 < 0, \quad y = 6 > 0$$

✔ Correct Option: (B) $x < 0, y > 0$

Question 65

The number of integer solutions of the equation $|1-i|^x = 2^x$ is

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Options:

- A. 1
- B. 0
- C. 2
- D. 3

Answer: A

Solution:

$$|1 - i|^x = 2^x$$

$$\Rightarrow \left(\sqrt{1^2 + (-1)^2} \right)^x = 2^x$$

$$\Rightarrow (\sqrt{2})^x = 2^x$$

$$\Rightarrow 2^{\frac{x}{2}} = 2^x$$

$$\Rightarrow \frac{x}{2} = x$$

$$\Rightarrow 0 = \frac{x}{2}$$

$$\Rightarrow x = 0$$

Number of integer solution is 1.

Question66

Let Z_1, Z_2 and Z_3 be three non zero complex numbers such that $a = |Z_1|, b = |Z_2|$

and $c = |Z_3|$, if the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then

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Options:

- A. $|Z_1| = |Z_2| = |Z_3| = abc$
- B. $|Z_1| + |Z_2| + |Z_3| = 0$
- C. $|Z_1| + |Z_2| + |Z_3| = abc$
- D. $|Z_1 - Z_2| = |Z_2 - Z_3|$

Answer: B

Solution:

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = a(bc - a^2) - b(b^2 - ac) + c(ab - c^2)$$

$$\Rightarrow abc - a^3 - b^3 + abc + abc - c^3 = 0$$

$$\Rightarrow 3abc - a^3 - b^3 - c^3 = 0$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 0$$

$$\Rightarrow (a + b + c) = 0$$

$$\text{Since, } a = |Z_1|, b = |Z_2|, c = |Z_3|$$

$$\therefore |Z_1| + |Z_2| + |Z_3| = 0$$

Question 67

If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, where z_1 and z_2 are two complex numbers, then

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Options:

A. $\frac{z_1}{z_2}$ is purely real

B. $\frac{z_1}{z_2}$ is purely imaginary

C. $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{4}$

D. $\left|\frac{z_1}{z_2}\right| = 1$

Answer: B

Solution:

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \quad (\text{given}) \dots (i)$$

$$\text{Let } z_1 = x + iy, z_2 = a + ib.$$

Then

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{x + iy}{a + ib} = \frac{x + iy}{a + ib} \times \frac{a - ib}{a - ib} \\ \frac{z_1}{z_2} &= \frac{(ax + yb) + i(ay - xb)}{a^2 + b^2} \dots (ii) \end{aligned}$$

From Eq. (i), we get

$$\begin{aligned} \text{LHS} &= |z_1 + z_2|^2 = |x + a + i(y + b)|^2 \\ &= (x + a)^2 + (y + b)^2 \\ &= x^2 + a^2 + 2ax + y^2 + b^2 + 2by \\ \text{RHS} &= |x + iy|^2 + |a + ib|^2 \\ &= x^2 + y^2 + a^2 + b^2 \\ \therefore \text{LHS} &= \text{RHS} \\ \Rightarrow x^2 + y^2 + a^2 + b^2 + 2(ax + by) \\ &= x^2 + y^2 + a^2 + b^2 \\ \Rightarrow 2(ax + by) &= 0 \Rightarrow ax + by = 0 \end{aligned}$$

Put $ax + by = 0$ in Eq. (ii),

$\frac{z_1}{z_2} = \frac{i(ay - xb)}{a^2 + b^2}$, which is purely imaginary.

Question 68

A real value of x will satisfy the equation, $\left(\frac{3-4ix}{3+4ix}\right) = \alpha - i\beta$, (α, β are real), if

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Options:

A. $\alpha^2 - \beta^2 = -1$

B. $\alpha^2 - \beta^2 = 1$

C. $\alpha^2 + \beta^2 = 1$

D. $\alpha^2 - \beta^2 = 2$

Answer: C

Solution:

$$\begin{aligned}\text{LHS} &= \frac{3-4ix}{3+4ix} \times \frac{3-4ix}{3-4ix} \\&= \frac{9+16i^2x^2-24ix}{9-(16i^2x^2)} = \frac{9-16x^2-24ix}{9+16x^2} \\&= \frac{9-16x^2}{9+16x^2} - \frac{24ix}{9+16x^2} = \alpha - i\beta = \text{RHS} \\ \therefore \alpha &= \frac{9-16x^2}{9+16x^2} \\ \beta &= \frac{24x}{9+16x^2} \\ \Rightarrow a^2 + \beta^2 &= \frac{(9-16x^2)^2 + (24x)^2}{(9+16x^2)^2} \\&= \frac{(9+16x^2)^2}{(9+16x^2)^2} = 1 \\ \therefore \alpha^2 + \beta^2 &= 1\end{aligned}$$

Question 69

What is the value of $(1 - i\sqrt{3})^9$ is equal to

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Options:

A. 2^9

B. -2^9

C. $2^9 i$

D. $-2^9 i$

Answer: B

Solution:

$$\begin{aligned}(1 - i\sqrt{3})^9 &= 2^9 \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right)^9 \\&= 2^9 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^9 = 2^9 \left(e^{-i\pi/3} \right)^9 \\&= 2^9 \left(e^{-i3\pi} \right) = 2^9 (\cos 3\pi - i \sin 3\pi) \\&= 2^9 ((-1)^3 - 0) = -2^9\end{aligned}$$

Question 70

$\left(\frac{\sqrt{6}-\sqrt{2}}{4} + \frac{\sqrt{6}+\sqrt{2}}{4} i \right)^{2020}$ is equal to

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Options:

A. $\frac{1}{2} + \frac{\sqrt{3}}{2} i$

B. $\frac{-1}{2} + \frac{\sqrt{3}}{2} i$

C. $\frac{-1}{2} - \frac{\sqrt{3}}{2} i$

D. $\frac{1}{2} - \frac{\sqrt{3}}{2} i$

Answer: D

Solution:

$$\begin{aligned}&\left(\sqrt{2} \frac{(\sqrt{3}-1)}{2 \cdot 2} + i \sqrt{2} \frac{(\sqrt{3}+1)}{2 \cdot 2} \right)^{2020} \\&= \left(\frac{\left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)}{\sqrt{2}} + i \frac{\left(\frac{\sqrt{3}}{2} + \frac{1}{2} \right)}{\sqrt{2}} \right)^{2020} \\&= \left(\frac{\frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)}{1} + \frac{i \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} + \frac{1}{2} \right)}{1} \right)^{2020}\end{aligned}$$

$$\begin{aligned}
&= \left(\left(\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \right) + i \left(\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \left(\frac{1}{\sqrt{2}} \cdot \frac{1}{2} \right) \right) \right)^{2020} \\
&= (\cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ) + i(\sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ)^{2020} \\
&= (\cos 75^\circ + i \sin 75^\circ)^{2020} \\
&= \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)^{2020} \\
&= \left(\cos \frac{5\pi}{12} \times 2020 + i \sin \frac{5\pi}{12} \times 2020 \right) \text{ [De-moivre]} \\
&= \left(\cos \frac{2525\pi}{3} + i \sin \frac{2525\pi}{3} \right) \\
&= (\cos(2525 \times 60)^\circ + i \sin(2525 \times 60)^\circ) \\
&= (\cos(n\pi - 60)^\circ + i \sin(n\pi - 60)^\circ), \text{ where } n = 842 \\
&= ((-1)^n \cos 60^\circ + i(-\sin 60^\circ)) \\
&= (\cos 60^\circ - i \sin 60^\circ) [\because n = \text{even}] \\
&= \frac{1}{2} - i \frac{\sqrt{3}}{2}
\end{aligned}$$

Question71

If $z_1 = 2 + 3i$ and $z_2 = 3 + 2i$, where $i = \sqrt{-1}$, then $\begin{bmatrix} z_1 & z_2 \\ -\bar{z}_2 & \bar{z}_1 \end{bmatrix} \begin{bmatrix} \bar{z}_1 & -z_2 \\ \bar{z}_2 & z_1 \end{bmatrix}$ is equal to

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Options:

A. $13I$

B. I

C. $26I$

D. Zero matrix

Answer: C

Solution:

$$\begin{aligned}
&z_1 = 2 + 3i, z_2 = 3 + 2i \\
&\Rightarrow \begin{bmatrix} z_1 & z_2 \\ -\bar{z}_2 & \bar{z}_1 \end{bmatrix} \begin{bmatrix} \bar{z}_1 & -z_2 \\ \bar{z}_2 & z_1 \end{bmatrix} \\
&= \begin{bmatrix} |z_1|^2 + |z_2|^2 & 0 \\ 0 & |z_1|^2 + |z_2|^2 \end{bmatrix} \\
&\Rightarrow |z_1|^2 = 4 + 9 = 13 \\
&\Rightarrow |z_2|^2 = 9 + 4 = 13 = \begin{bmatrix} 26 & 0 \\ 0 & 26 \end{bmatrix} = 26I
\end{aligned}$$

Question72

The radius of the circle represented by $(1 + i)(1 + 3i)(1 + 7i) = x + iy$ is ($i = \sqrt{-1}$) .

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Options:

- A. 1000
- B. $10\sqrt{10}$
- C. 10000
- D. 100

Answer: B

Solution:

$$\begin{aligned} (1 + i)(1 + 3i)(1 + 7i) &= x + iy \\ \Rightarrow (-2 + 4i)(1 + 7i) &= x + iy \\ \Rightarrow -30 - 10i &= x + iy \\ \Rightarrow |x + iy| &= \sqrt{30^2 + 10^2} \\ &= \sqrt{1000} = 10\sqrt{10} \end{aligned}$$

Question73

If $1, \alpha_1, \alpha_2, \alpha_3$ and α_4 are the roots of $z^5 - 1 = 0$ and ω is a cube root of unity, then $(\omega - 1)(\omega - \alpha_1)(\omega - \alpha_2)(\omega - \alpha_3)(\omega - \alpha_4) + \omega$ is equal to

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Options:

- A. 0
- B. -1
- C. -2
- D. 1

Answer: C

Solution:

$\because 1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the roots of $z^5 - 1 = 0$

$$\begin{aligned}\Rightarrow & (z-1)(z-\alpha_1)(z-\alpha_2)(z-\alpha_3)(z-\alpha_4) = z^5 - 1 \\ \Rightarrow & (\omega-1)(\omega-\alpha_1)(\omega-\alpha_2)(\omega-\alpha_3)(\omega-\alpha_4) = \omega^5 - 1 \\ & (\omega-1)(\omega-\alpha_1)(\omega-\alpha_2)(\omega-\alpha_3)(\omega-\alpha_4) + \omega \\ & = (\omega^5 - 1) + \omega = \omega^2 - 1 + \omega \\ & = -1 - 1 = -2\end{aligned}$$

Question 74

If $a > 0$ and $z = x + iy$, then $\log_{\cos^2 \theta} |z - a| > \log_{\cos^2 \theta} |z - ai|$, ($\theta \in R$) implies

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Options:

- A. $x > y$
- B. $x < y$
- C. $x + y = \cos \theta$
- D. $x + y < 0$

Answer: A

Solution:

$$\begin{aligned}a &> 0, z = x + iy \\ \log_{\cos^2 \theta} |z - a| &> \log_{\cos^2 \theta} |z - ai| \\ 0 &< \cos^2 \theta < 1 \\ \text{So, } |z - a| &< |z - ai| \\ \Rightarrow (x - a)^2 + y^2 &< x^2 + (y - a)^2 \\ \Rightarrow x^2 + a^2 - 2ax + y^2 &< x^2 + y^2 + a^2 - 2ay \\ \Rightarrow -2ax &< -2ay \\ \Rightarrow x &> y\end{aligned}$$

Question 75

If one root of the equation $ix^2 - 2(i+1)x + (2-i) = 0$ is $(2-i)$, then the other root is

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Options:

- A. $-i$

B. $2 + i$

C. i

D. $2 - i$

Answer: A

Solution:

$$ix^2 - 2(i + 1)x + (2 - i) = 0 \dots (i)$$

Let the other roots be α .

$$[x - (2 - i)][x - \alpha] = 0$$

$$\Rightarrow x^2 - (2 - i + \alpha)x - (\alpha(2 - i)) = 0$$

{comparing with Eq. (i)}

$$\Rightarrow ix^2 - (2i + 1 + \alpha i)x - [\alpha(2i + 1)] = 0$$

$$\Rightarrow 2i + 1 + \alpha i = 2i + 2$$

$$\Rightarrow$$

$$\alpha i = 1$$

$$\Rightarrow \alpha = \frac{1}{i} = -i$$

Question76

If $|z - 2| = |z - 1|$, where z is a complex number, then locus z is a straight line

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Options:

A. Parallel to X - axis

B. Parallel to Y -axis

C. Parallel to $y = x$

D. Parallel to $y = -x$

Answer: B

Solution:

$$\because |z - 2| = |z - 1|$$

Let $z = x + iy$, then

$$\Rightarrow |x + iy - 2|^2 = |x + iy - 1|^2$$

$$\Rightarrow (x - 2)^2 + y^2 = (x - 1)^2 + y^2$$

$$\Rightarrow x^2 - 4x + 4 = x^2 - 2x + 1$$

$$\Rightarrow -2x + 4 = 4 - 1 \Rightarrow x = 3/2$$

Which is parallel to Y -axis.

Question77

If $\left(\frac{1+i}{1-i}\right)^m = 1$, then m cannot be equal to

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Options:

A. 1934

B. 2024

C. 2172

D. 10^{100}

Answer: A

Solution:

$$\left(\frac{1+i}{1-i}\right)^m = 1 \Rightarrow \left[\frac{(1+i)(1+i)}{(1-i)(1+i)}\right]^m = 1$$

$$\Rightarrow \left(\frac{1-1+2i}{1+1}\right) = 1$$

$$\Rightarrow i^m = 1$$

$$\Rightarrow m \text{ is multiple of } 4 \left[\cdot \cdot i^{4n} = 1, \forall n \in N \right]$$

Option (a) is not a multiple of 4 .

$$\therefore m \neq 1934$$

Question78

$(\sin \theta - i \cos \theta)^3$ is equal to

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Options:

A. $i^3(\cos 3\theta + i \sin 3\theta)$

B. $\cos 3\theta + i \sin 3\theta$

C. $\sin 3\theta - i \cos 3\theta$

D. $(-i)^3(\cos 3\theta + i \sin 3\theta)$

Answer: D

Solution:

$$\begin{aligned}(\sin \theta - i \cos \theta)^3 &= (-i)^3(\cos \theta + i \sin \theta)^3 \\ &= (-i)^3(\cos 3\theta + i \sin 3\theta) \text{ [by De-Moivre theorem]}\end{aligned}$$

Question 79

Real part of $(\cos 4 + i \sin 4 + 1)^{2020}$ is

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Options:

- A. $2^{2020} \cos^{2020} 2 \cos 2020$
- B. $2^{2020} \cos^{2020} 2 \cos 4040$
- C. $2^{1020} \cos^{2020} 2 \cos 4040$
- D. $2^{2020} \cos^{2020} 1 \cos 2020$

Answer: B

Solution:

$$\begin{aligned}\text{Real part of } (\cos 4 + i \sin 4 + 1)^{2020} &= \text{Real part of } (2 \cos^2 2 + i 2 \sin 2 \cos 2)^{2020} \\ &= \text{Real part of } 2^{2020} \cos^{2020} (2)(\cos 2 + i \sin 2)^{2020} \\ &= \text{Real part of } 2^{2020} \cos^{2020} (2)(\cos 4040 + i \sin 4040) \\ &= 2^{2020} \cos^{2020} 2 \cdot \cos 4040 \quad [z + \bar{z} = 2 \operatorname{Re}(z)]\end{aligned}$$
